Proc. Instn Civ. Engrs, Part 1, 1989, **86**, Aug., 709–770
9430 TRANSPORTATION ENGINEERING GROUP

Singapore Mass Rapid Transit System: construction

T.W. HULME, BSc, ARSM, FICE, FIStructE, MIMM, MASCE*

L. A. C. POTTER, FIStructE, MIHT*

J. N. SHIRLAW, BSc, MICE, MHKIE, FGS*

The Paper is introduced by a brief description of the regulation of construction in Singapore. Reference is made to the applicable legislation and its enforcement. Ground conditions are reviewed in the context of their influence on the choice of construction methods and equipment. The principal features of underground station construction are given. The tunnel works are dealt with in some detail because these are of special interest in the densely developed central area of Singapore. Cut-and-cover tunnelling used various support methods with both conventional and underwater excavation. Bored tunnelling embraced full face mechanized machines at one extreme and Greathead shields with hand excavation at the other. A comparison and analysis of the results is given. New Austrian tunnelling method variants are described. Tunnelling aids are described in general terms and details given of instances of special interest. The use of compressed air and particularly its medical supervision is referred to. The specified construction method for viaducts is described and the various launching gantry types developed by contractors described. Casting yard layouts are given and the principal conclusions derived from the Singapore Mass Rapid Transit Corporation's experience in these aspects are given. The main elevated station types are described and the various construction advantages and disadvantages given. The three depots posed significant foundation problems and the contrasting construction techniques required are detailed. Track-laying techniques and rates are described for the three types of permanent way used.

Introduction

For construction purposes the Singapore Mass Rapid Transit System (MRT) was divided into five phases, as shown in Fig. 1. It was originally intended to construct the various phases in sequence, so as not to impose an excessive demand on the available construction resources in Singapore. However, a severe contraction of construction demand during the 1984–85 recession in Singapore prompted the programme to be accelerated so that, for a period in 1987, all the 42 stations were either under design, construction, or final fitting out and commissioning. The speed and scale of the works were large: during the tunnelling of phase 1 there were 18 separate 6 m dia. tunnel faces being worked simultaneously; the completion of 47 route km of viaduct in phases 1A, 2A and 2B was achieved in three years; 190 km of track were laid in three working years.

Ordinary meeting 3.30 p.m., 14 November 1989. Written discussion closes 28 November 1989; for further details see p. ii.

* Mass Rapid Transit Corporation, Singapore.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

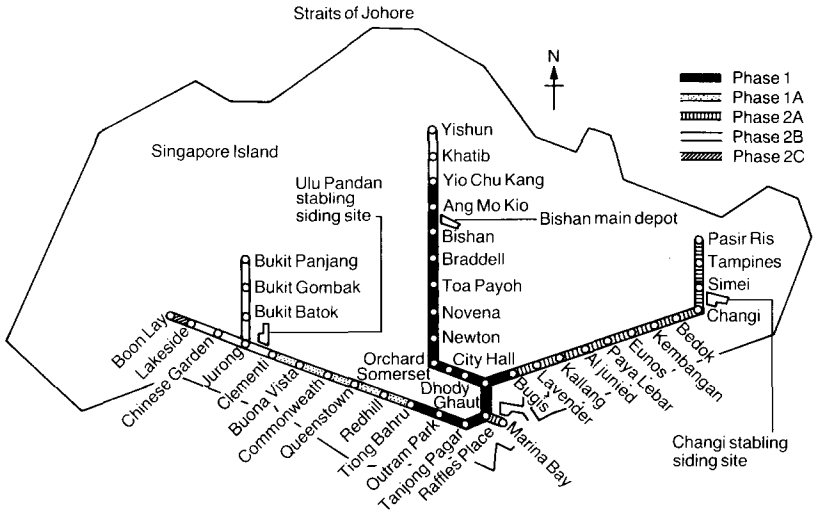


Fig. 1. Singapore MRT System showing construction phases

2. The ground breaking ceremony for phase 1 was held on 22 October 1983, with five stations becoming operational on 7 November 1987, and the whole of phase 1 on 12 December 1987. Phase 1A opened on 12 March 1988 and the rest of the system is expected to be operational by mid-1990. At the time of publication the major civil construction work is basically complete.

3. To achieve a rapid start to the work phase 1 was let mainly on a design-and-construct basis, and was divided into 14 major civil contracts. The subsequent phases were designed by architectural/engineer (A/E) consultants appointed by the Mass Rapid Transit Corporation (MRTC). Construction of phases 1A to 2C were let under a further 21 major civil contracts. Three of the civil contracts were to build the depots at Bishan, Ulu Pandan and Changi. The underground contracts generally covered a shorter route length than the viaduct contracts, because of the higher unit value. The underground contracts varied from a single station to, at most, two stations and their connecting tunnels. Overhead contracts typically included two stations, occasionally three, with their connecting viaducts.

4. The scale of the works made it inevitable that a significant proportion of the major contracts, particularly in the underground sections, would go to overseas rather than Singaporean contractors and consultants. Among the various contractors and consultants who were successful tenderers, there were representatives from Japan, Hong Kong, Belgium, UK, USA, France, Switzerland, Italy, Korea and Taiwan. Overseas contractors were encouraged, by use of a preferential margin on tenders, to form joint ventures with local contractors. Technology transfer both to local employees of foreign firms and to local companies was actively promoted. The many expatriate staff working on the project had to adapt to the local construction regulations, which in some instances were significantly different from those applying in their home countries.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Singapore construction regulations

5. The regulation of construction of public works in Singapore is governed by four principal pieces of legislation

(*a*) the Building Control (Administration) Act (1979), the Building Control (Construction) Regulations (1979) and Building Control (Space, Light and Ventilation) Regulations (1979) made thereunder; these will be superseded by the Building Control Bill, currently before Parliament.
(*b*) the Factories Act (1973) and the Factories (Building Operations and Works of Engineering Construction) Regulations (1985) made thereunder.
(*c*) the Professional Engineers Act (1970)
(*d*) the Architects Act (1976).

6. The first act sets out the procedures to be observed in starting and carrying out building works. The second resembles the UK Factories Act in that it defines the law relating to the health, safety and welfare of persons employed in factories and other places of work. It governs the conduct of civil construction works and the actions of individuals on site, and more particularly lays down the responsibilities of the parties concerned for health and safety and prescribes the penalties for non-observance.

7. The Professional Engineers Act is of particular significance and departs from UK practice. It defines the responsibilities of the registered professional engineers as regards both design and construction. The legal responsibility of the engineer is personal and action against the individual is taken in the event that the works are not constructed in accordance with the drawings and specifications. Similar legislation governs the practice of architects.

8. In the context of the Singapore MRT project these acts had significant weaknesses. The supervision of the works, as commonly understood, was undertaken by the MRTC with directly employed engineers and inspection staff. However, to preserve the legal responsibility of the work outwith the MRTC's staff it was decided to require the contractor to undertake the responsibilities implicit in the Professional Engineers Act. For that, Clause 2(2) in the FIDIC Conditions of Contract was revised to read 'The functions of the Engineers Representative are to watch and *inspect* the construction completion and maintenance of the Works ...' The original clause read: 'The Engineers Representative shall be responsible to the Engineer and his duties are to watch and *supervise* the Works ...'.

9. In the A/E designed contracts a waiver was sought from the Professional Engineers Board to separate the professional responsibility of certifying the design and the supervision of construction. The latter was then specified as part of the contractor's duties.

10. The demarcation between legal responsibility for the supervision of construction, which was laid on the contractor, and the observance of specified requirements of the MRTC, which were enforced by the MRTC's supervisory staff, was relatively straightforward. The legal duty to construct in accordance with the employer's drawings and specifications and variations instructed by authorized persons, is in fact no more than a responsible contractor ordinarily undertakes.

11. A number of construction aspects of the project were new to Singapore. In particular compressed air working, although previously sparingly used, was unregulated. With the significant volume of work likely to be carried out in this

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

manner, and at the pressures required (up to 250 kPa), this situation was obviously unacceptable. The Ministry of Labour undertook a comprehensive review of existing legislation worldwide and also arranged visits for its officials overseas.

12. Meanwhile it was necessary for the MRTC to specify suitable standards in the MRTC's contracts until the necessary legislation could be enacted. The legislation in Hong Kong was based largely on UK practice, was familiar to many of the contractors likely to tender—Japanese, European, American and Australian—and was considered as up to date as any. This was therefore adopted in the MRTC's requirements. The contractors were also required to observe the good working practices recommended in CIRIA report 44, *Medical code of practice for work in compressed air*.

13. Additional payment to cover differences between the specified practice and that subsequently enacted by the Singapore Parliament was specifically excluded in the conditions of contract. In the event the Singapore legislation was basically compatible with the specified requirements. A minimum diameter (1·5 m) was, however, required for boiler type person locks.

14. Inspection of metallic air locks during manufacture to ensure their compliance with the relevant pressure vessel regulations—including an initial request for water testing—caused some confusion but the Ministry of Labour after discussion modified their procedures. The benefit of a fresh and clinically efficient approach exposed a number of preconceived ideas based on overseas practice. More independent inspection and regulation of air lock fabrication, if expected, adds virtually nothing to cost but may avoid serious failure in operation and is recommended.

15. Other relevant legislation controlling specific aspects of construction is

(*a*) environment: the Water Pollution Control and Drainage Act (1975); the Sanitary Plumbing and Drainage System Regulations (1976) and code of practice made thereunder; the Surface Water Drainage Regulations (1976) and code of practice made thereunder; the Environmental Public Health Act (1970); the Trade Effluent Regulation (1976)
(*b*) road works: the Local Government (Private Streets) Regulation; the Local Government Integration Act (1985)
(*c*) utilities: the Public Utilities Act (1970)
(*d*) fire regulations: the Fire Precautions in Buildings Act (1982)
(*e*) planning: the Planning Act (1970)

The construction of the whole of the works was authorized by the Mass Rapid Transit Corporation Act (1983).

MRT staff organization

16. The duties of the MRTC's supervisory staff were essentially to assure the correct construction and measurement of the works, certify payment, administer the contract including claims and variations, etc. The organization was thus conventional. The works were divided into geographically defined sections each under the overall control of a construction manager reporting to the project manager. The latter was designated the Engineer for all the major contracts, but all the Engineer's powers except those relating to claims and extension of time were delegated to the construction managers. These two items were retained by the Engineer to ensure uniformity of application of these often contentious clauses.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Below the construction manager a hierarchy of senior resident engineers, resident engineers, engineers, inspectors and specialists—geotechnical engineers, surveyors, planners, etc., was implemented as shown in Fig. 2. At the height of the construction there were six construction managers and approximately 900 site supervisory staff. The contractor's workforce totalled 12 000 at its peak when 35 major contracts were active.

17. Support to the site staff was provided by the survey, contract administration and safety departments. In addition there were design departments for both the civil and architectural works. Although little of the actual design was carried out by these departments, they were responsible for checking the contractor's temporary works designs, vetting the permanent works design and coordination between the various consultants and contractors.

Construction methods

18. To cover fully all of the construction methods of a S$5000 million railway system would be impossible within the constraints of this paper. Only a selection of topics can be covered. These will be

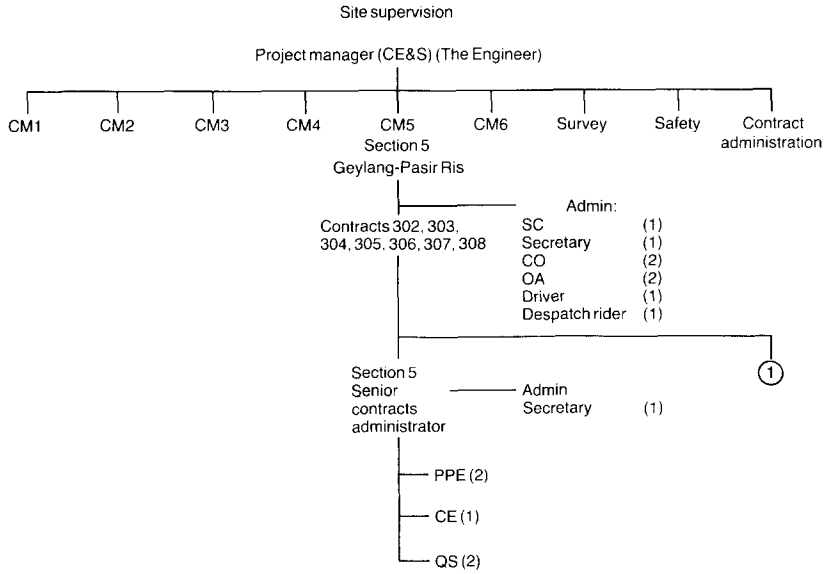
(*a*) underground works: methods of temporary works and tunnel excavation, soil improvement and underpinning, settlements associated with the work
(*b*) depots: methods of site formation
(*c*) viaducts: different beam launching methods, construction of foundations, piers and crossheads, and elevated stations
(*d*) track laying.

Engineering geology

19. The geology of Singapore can be conveniently divided into the predominantly soft clays and loose sands of the Kallang formation, and the earlier, generally stronger, soils and rocks. The Kallang formation is recent, of Holocene and late Pleistocene age, and has been laid down in valleys eroded into the underlying rocks during periods of low sea levels. The most common material within the Kallang formation is a normally to lightly overconsolidated silty clay of marine origin. A shear strength depth profile of a deep marine clay deposit at Bugis station is shown in Fig. 3. Lenses of fluvial sand and clay occur within the marine clay particularly at the interface of an older, late pleistocene member and a younger, holocene member. The marine clay has a low permeability, approximately 10^{-9} m/s, but is compressible with a typical C_c of 0·85. Inland the soft clay tends to have a higher organic content, probably because it was laid down under estuarine or marshy conditions. This estuarine clay is even more compressible than the marine clay, with C_c values of 1–5.

20. The main rock types in Singapore are the igneous rocks of the Bukit Timah granite and Gombak Norite formations, and the sedimentary rocks of the Jurong formation. Bukit Timah granite is used to describe a suite of granitic rocks and is more typically a grandiorite than a true granite. Both the igneous and sedimentary rocks show varying depths and degrees of weathering. Completely weathered rocks of these formations tend to be of low permeability and to outwardly resemble stiff clays. The weathering grade classification used on MRT work was a modified version of that proposed by the Geological Society of London. The modification was to simplify the classifications so that the normally

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August



CM	construction manager	AQS	assistant quantity surveyor
SC	senior clerk	PPE	programming/planning engineer
CO	clerical officer	GE	geotechnical engineer
OA	office attendant	SRE	senior resident engineer
STO(S)	senior technical officer (survey)	RE	resident engineer
TO(S)	technical officer (survey)	CE	contracts engineer
T(S)	technician (survey)	ARE	assistant resident engineer
C(S)	chairman (survey)	SE	site engineer
TO(A)	technical officer (architectural)	SIOW	senior inspector of works
SQS	senior quantity surveyor	WS	works supervisor
QS	quantity surveyor		

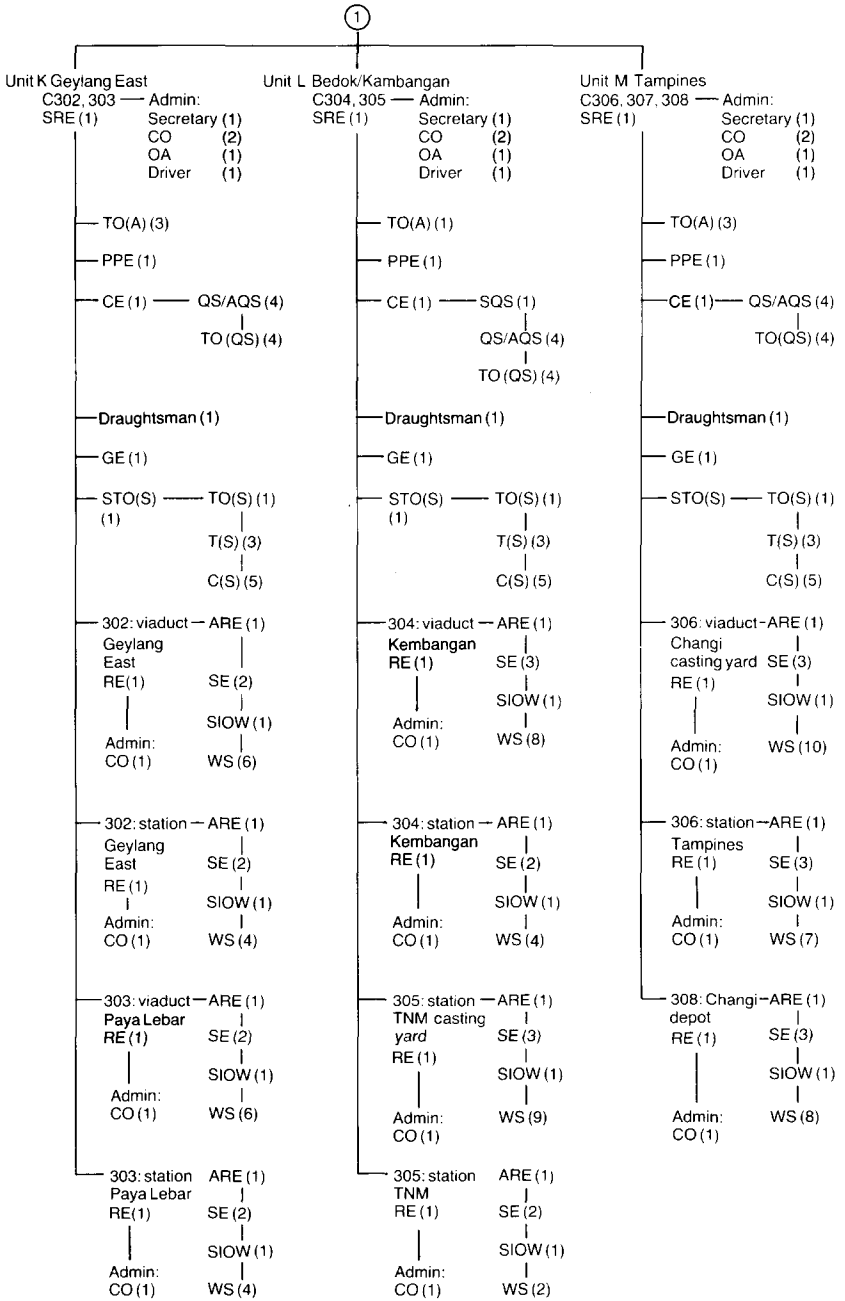
Fig. 2. Site supervision

accepted six grades were reduced to three (Table 1). In addition a category 3 was introduced which would normally be covered by a rock mass description rather than a weathering grade. The modified system did cause some confusion, and Poh *et al.*[1] recommended that it be discontinued. However, the modified system is still used in this Paper for convenience.

21. Two other formations, the Old Alluvium and the Singapore Boulder Bed (S3), of later tertiary or early Quatenary age, also had a significant influence on construction. The Old Alluvium is a partially lithified, dense deposit of interbedded sands and clays. The Singapore Boulder Bed, which underlies much of the central business district, was originally thought to be part of the weathering sequence of the Jurong formation (hence S3), but is now thought to be colluvial in origin.[1] It consists of often massive, strong sandstone boulders in a stiff clay matrix.

22. Particularly in the underground works, the nature and variety of the geology encountered was the single most important factor in determining the

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August



Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

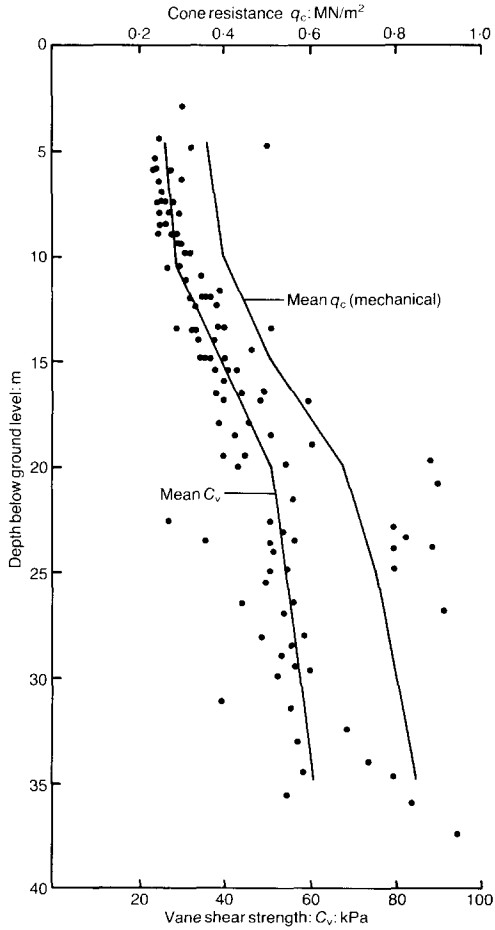


Fig. 3. Shear strength depth profile of marine clay at Bugis station

methods used in construction. Fig. 4 shows the ground encountered and the variety of techniques used in just 2·3 km of the underground route. This section shows just over 10% of the underground route, and less than 4% of the total route.

23. Of the various geological formations encountered, the presence or absence of the Kallang formation was usually a critical factor in the choice of construction method. In the underground sections the high compressibility and low shear strength of the Kallang formation could quickly result in large settlements unless the work was carefully planned and executed. In the overhead (viaduct) sections not only did the foundations have to extend through the soft deposits, but allowance had to be made for possible downdrag (negative skin friction).

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/ficep_1989_2628.pdf by National University of Singapore user on 18 August

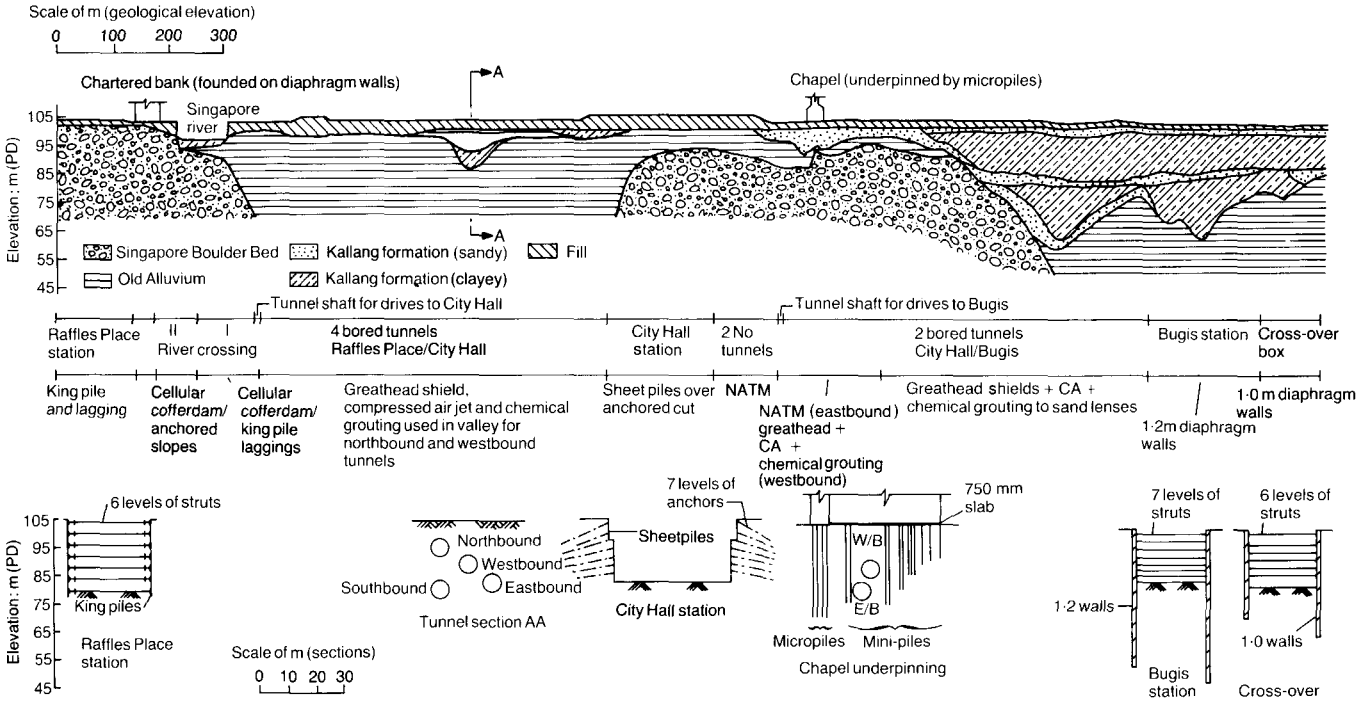


Fig. 4. Section showing geology of contracts 107A, 107, 107B and 301

Table 1. Modified weathering grade classification for Singapore rocks (after Dames and Moore)

Symbol*	General description
G1, S1	Fresh to slightly weathered rock (weathering grades 1 and 2)
G2, S2	Moderately to highly weathered rock (weathering grades 3 and 4)
G3, S3	Bouldery soil: boulders of rock of variable weathering within completely weathered rock or residual soil
G4, S4	Completely weathered rock or residual soil (weathering grades 5 and 6)

* G = Bukit Timah Granite; S = Jurong Formation sedimentary rock.

Underground work

Cut and cover construction: stations

24. Effective planning procedures and the generous open areas within Singapore meant that suitable sites for cut-and-cover construction were provided for all the underground stations. In most cases the sites were off street, avoiding costly traffic and utility diversions. In three cases, Newton, Somerset and Lavender stations, roads were closed and diversions provided around the site. At three other stations, City Hall, Bugis and Marina Bay, temporary decks were provided to carry traffic over the excavations. The temporary decks were moved around to suit the construction sequence, which in the case of Marina Bay meant repeatedly moving a six-lane expressway.

25. Support systems during excavation were chosen and designed by the contractors. The major factors constraining the choice of system were geology, depth of excavation, and the presence of buildings or roads near the excavation. The support systems are summarized in Table 2.

26. Where the Kallang formation was virtually absent and clear area available, cut slopes were used. The stability of the slopes was improved by soil nailing, rock-bolting or anchoring. Examples of cut-slope excavations at Tanjong Pagar and at Orchard stations are shown in Figs 5 and 6. At Tanjong Pagar the station was excavated in weathered Jurong formation, and it was originally intended to use rock-bolts only in the steepest (1 : 0·3) section of slope. However, a major incipient failure detected during excavation resulted in more substantial support being provided as shown in Fig. 5. At Orchard the station was excavated in completely weathered granite, and soil nails were installed to stabilize the G4 slope. Again, an incipient failure was detected, and the initial design was amended by lengthening the nails and installing horizontal drains (Fig. 6).

27. Major problems, affecting Singapore's St. Andrews Cathdral, also occurred at City Hall station (Fig. 7). The original design was for a 15 m high vertical cut face, supported by anchors, in type S3. Above the main cut a further 6 m of sand fill was supported by a sheet-piled wall. The problems included two

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Table 2. Temporary work for underground stations

Station	Maximum depth of excavation: m	Typical soil sequence*	Retaining system used	Special measures
Braddell	14·9	1F, G4	0·6 and 0·8 m diaphragm walls	
Toa Payoh	13·5	4F, 4K, G	Sheet piles	
Novena	14·7	½F, 14·2K, G	Sheet piles	
Newton	14·3	3F, 13K, G	0·8 m diaphragm walls	Jet grouting
Orchard	21	½F, G	Nailed slopes	
Somerset	16·2	2F, 8K, G	0·6 m diaphragm walls or sheet piles	
Dhoby Ghaut	16·1	1F, 10K, S	Sheet piles	
City Hall	22·3	3F, 2K, S3	King piles and shotcrete lagging	
Tanjong Pagar	17·9	½F, S	Slopes, anchors	
Outram	13·9	2F, 3K, S	8 m deep sheet piles over king piles and timber laggings	
Tiong Bahru	14·1	1F, S	King piles and shotcrete lagging	
Bugis	18·3	1F, 34K, O	1·2/1·0 m diaphragm walls	Lime piles
Lavender	16·5	3F, 20K, O	1·0 m/diaphragm walls	
Marina Bay	16·4	12F, 24K, O	Composite H pile/sheet pile	Underwater excavation

* F = fill
G = Granite, S3 = Jurong } including weathered rock
K = Kallang
S = Boulder Bed
O = Old Alluvium

Example: 3F, 13K, G = 3 m of fill overlying 13 m of Kallang deposit overlying granite (in this case completely weathered granite).

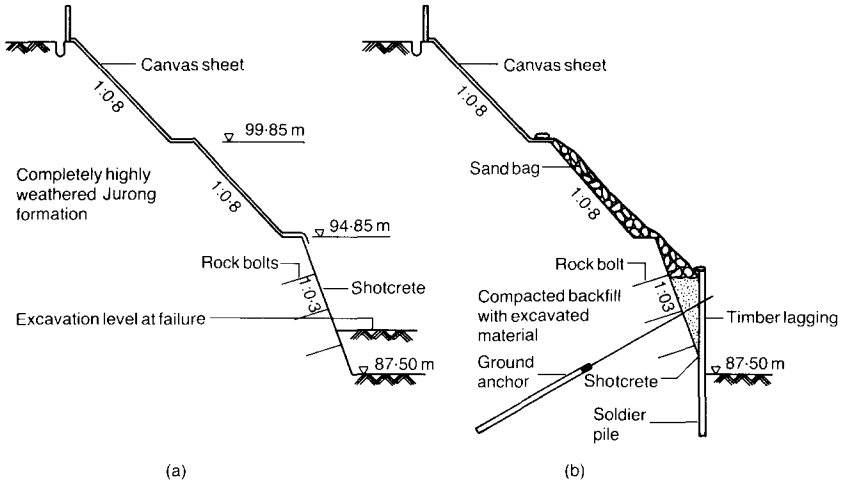


Fig. 5. Cut slopes at Tanjong Pagar station: (a) original design and (b) revised design due to incipient failure

failures at the face of the cut and settlement of the cathedral. Remedial measures, including temporary backfilling, a mini-pile wall, further anchors and struts, allowed satisfactory completion without further effect on the cathedral.

28. The repeated cases of major incipient failures reflect the difficulty of designing such slopes and cuts in weathered rocks, and the pressure for eco-

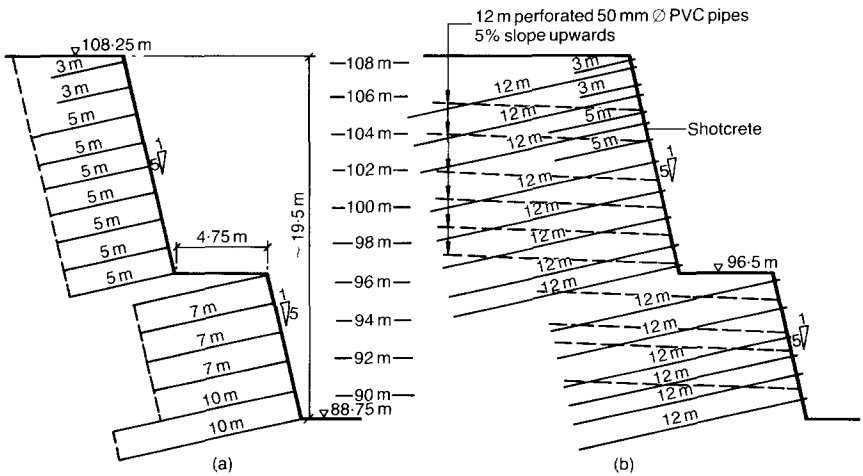


Fig. 6. Nailed slopes at Orchard station: (a) original design, nails at 1·5 m vertical spacing and 2 m horizontal spacing, front face protected by shotcrete; (b) revised design after failure detected when excavation reached + 100 m PD

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

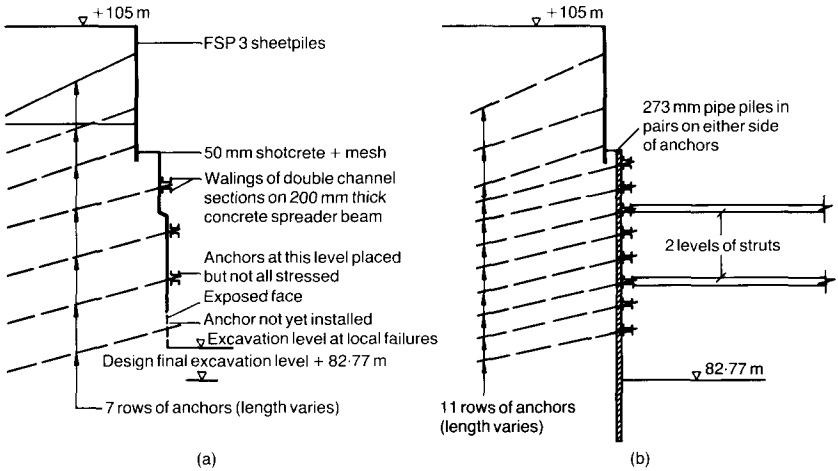


Fig. 7. Anchored cut at City Hall station: (a) original and (b) final designs

nomical design. In each case the incipient failure was identified by instrumentation, and suitable measures taken to stabilize the situation before there were any major adverse effects.

29. Another method of cut-and-cover support used in weathered rock was to install king piles and excavate providing additional support with struts and laggings. Strut design was generally based on the apparent earth pressure method suggested by Peck.[2] The appropriate coefficient to use in weathered rock was sometimes underestimated, and additional struts had to be installed when the strut monitoring showed excessive loads. Fig. 8 shows the back-calculated apparent earth pressure diagram for Raffles Place station.

30. Where there was any significant depth of the soft clays and loose sands of the Kallang formation, a continuous walling system was used. The stability number N of the marine clay reached 6 at a depth of 6–8 m, at which point it would extrude into any openings. Sheet piles were the most common type of walling installed, although thin (0·6 or 0·8 m) diaphragm walls were also used, particularly where it was necessary to minimize the effects of construction on nearby buildings.

31. A typical 16 m deep station excavation in marine clay would generate a net active pressure in marine clay until the shear strength (measured by field vane) exceeded about 60 kPa. As can be seen from Fig. 3, Singapore marine clay would not provide any net passive pressure shallower than a depth of 35 m. In most cases firmer strata where the sheet piles or diaphragm walls could obtain toe restraint were reasonably close to the base of the excavation. However, in two cases where the marine clay extended to 40 m below ground level, special construction measures had to be adopted because of the massive moments and deflexions that would otherwise have been induced in the support system.

32. At Bugis Station 1·2 m thick diaphragm walls were socketed 9–14 m into dense, cemented Old Alluvium, giving a maximum wall depth of 54 m. Where the depth to the Old Alluvium was greatest the marine clay just above and below the

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

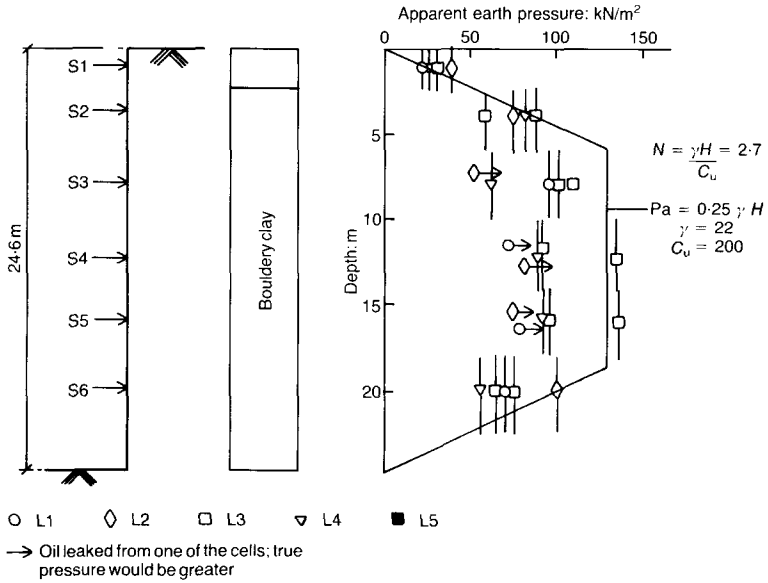


Fig. 8. Back-calculated Peck diagram for Raffles Place station

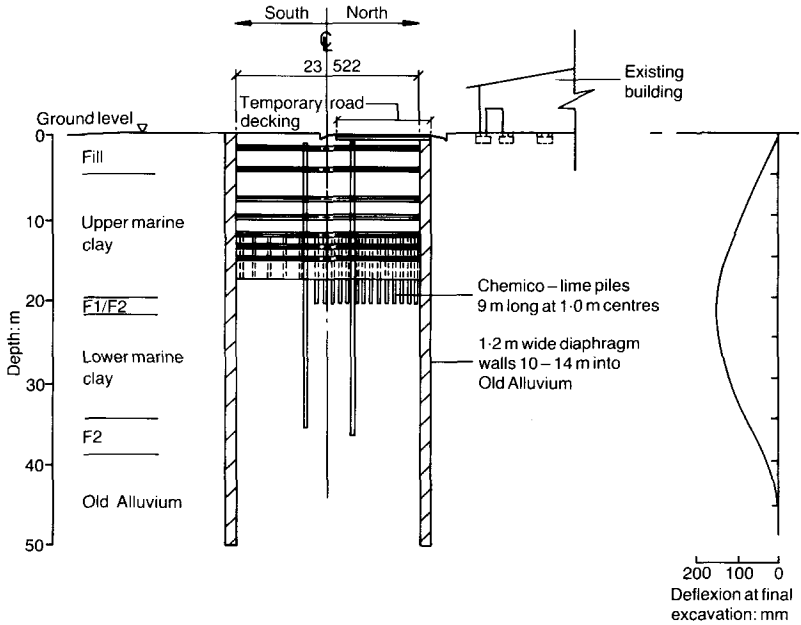


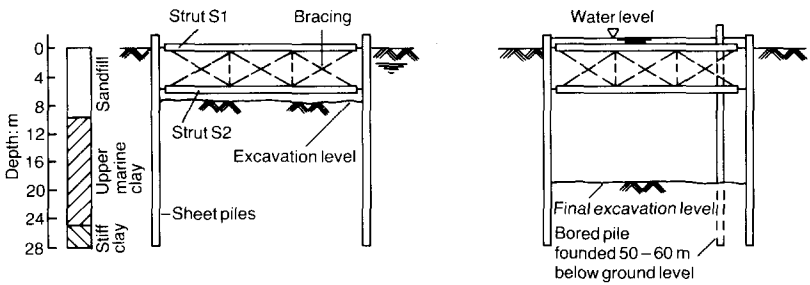
Fig. 9. Section at Bugis station

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

final excavation level was treated by chemico-lime piles (see Fig. 9) to restrict deflexion. Despite all these measures, and seven levels of strutting, deflexions of up to 150 mm were still recorded in the wall.

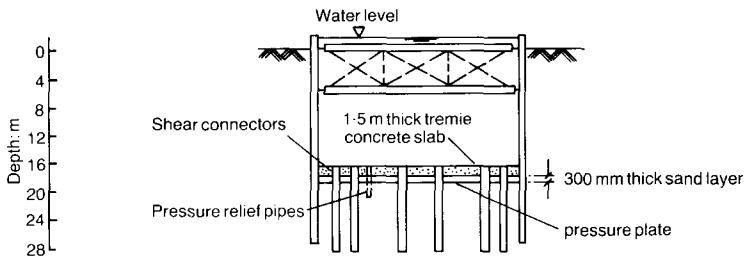
33. At Marina Bay station underwater excavation was used to help resist the active pressures. 25–30 m long composite H-pile/sheet pile sections were used, which stopped within the Kallang formation (Fig. 10). Approximately 7 m of dry excavation was possible without overstressing the wall. This excavation was used to install two levels of strutting to restrain the top of the wall. The cofferdam was then flooded and excavated underwater. When the final excavation level was reached the 1 m dia. bored piles designed to act as the station foundation were installed. A 1·5 m thick concrete slab was then poured using the tremie method, before the excavation was dewatered. This unreinforced slab acted a bottom strut and, anchored down by the piles, resisted base heave pressures. One of the great advantages of this scheme was the relatively unrestricted space provided to cast the permanent works (Fig. 11).

34. For the stations with diaphragm walls, and for Marina Bay station, simple calculation methods, such as the Peck diagram used elsewhere, were not appro-



Stage 1:
1. Drive sheet piles
2. Excavate about 1·5 m and install S1
3. Excavate about 6·5 m and install S2 and bracing between S1 and S2

Stage 2:
1. Flood the cofferdam to the top level
2. Excavate under water using grabs, water jets and air lifting
3. Install bored piles using RCD method



Stage 3:
1. Place 300 mm thick (minimum) sand levelling layer
2. Install pressure relief pipes
3. Install pressure plates
4. Cast tremie concrete slab

Fig. 10. Construction sequence for Marina Bay station

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

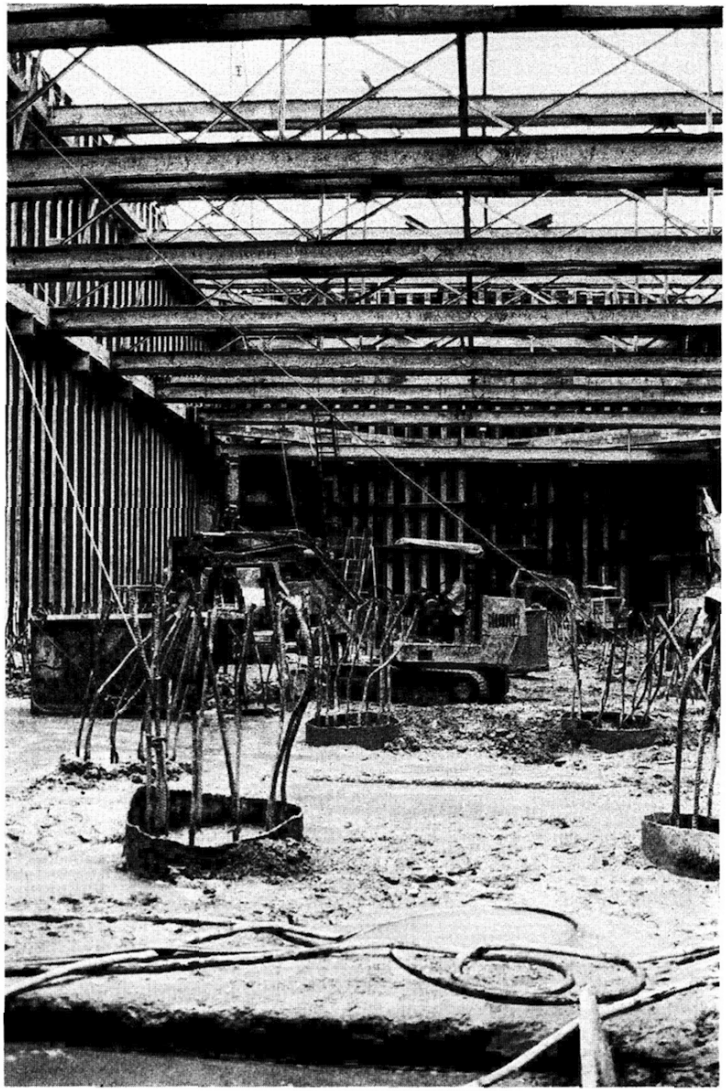


Fig. 11. Clear working space at Marina Bay station

priate, and computers had to be used in the analysis. The programs typically modelled the walls as beams on elasto-plastic springs, and the predictions were generally found to match the field measurements within a reasonable margin. The estimation of the uplift pressure on the base slab at Marina Bay was done by hand calculation and proved reasonably consistent with the readings on total-stress pressure cells placed under the slab (Fig. 12).

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August 20

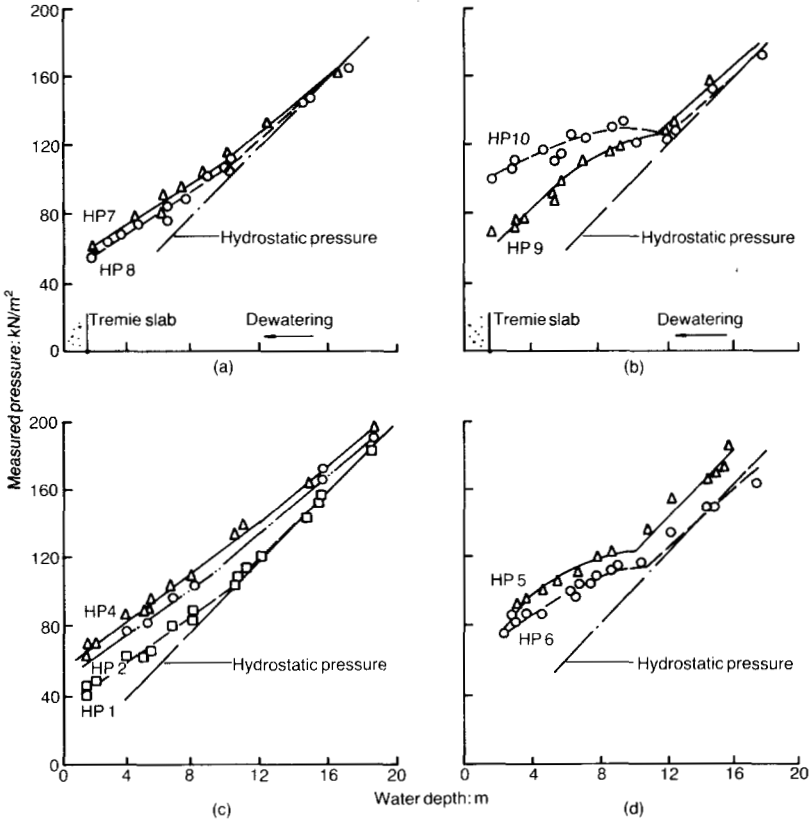


Fig. 12. Uplift pressures on Marina Bay station tremie slab: (a) pressure plates HP 7 and 8, cofferdam M2, predicted pressure 72 kN/m²; (b) pressure plates HP 9 and 10, cofferdam J, predicted pressure 83 kN/m²; (c) pressure plates HP 1, 2 and 4, cofferdam F, predicted pressure 83 kN/m²; (d) pressure plates HP 5 and 6, cofferdam A, predicted pressure 85 kN/m²

Cut-and-cover construction: tunnels

35. The tunnelled sections of the MRT were mostly within the highly developed central business district and the residential areas around it. Because of the concentration of roads, utilities and buildings on the surface, bored tunnelling was generally adopted as being both less disruptive and cheaper than cut-and-cover tunnelling. However, there were some locations where cut-and-cover construction was preferred. These fall into four categories.

36. *Underwater crossings.* The two underwater crossings, of the Singapore River and the Telok Ayer Basin, were both cut-and-cover tunnels. The Singapore River was crossed in two stages (Figs 13 and 14) with both stages using cellular cofferdams in the soft river bed sediments. Below the cellular cofferdams the main excavations for the tunnels were carried out using strutted king piles (north side)

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

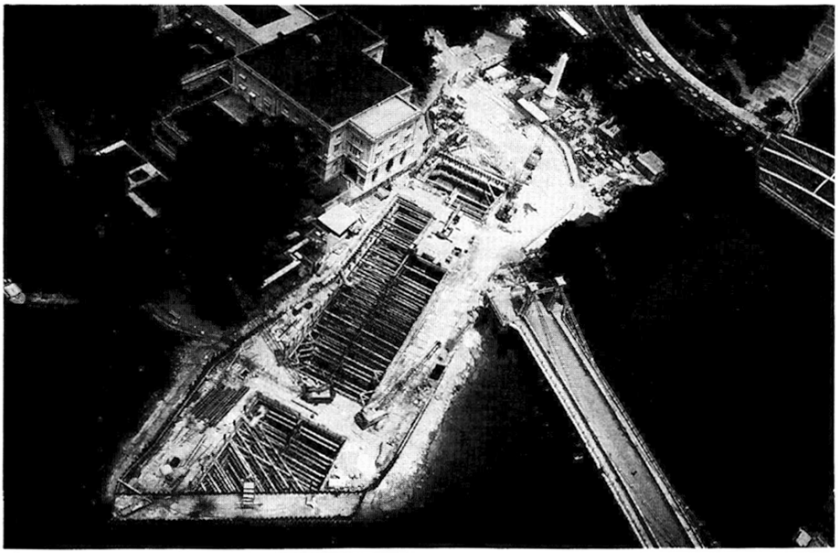


Fig. 13. River crossing, stage 1



Fig. 14. River crossing, stage 2

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

and cut slopes (the steepest being anchored) on the south side. The Telok Ayer Basin crossing was carried out using the same underwater excavation method as for the Marina Bay station.

37. *Cross-overs.* Underground railways require cross-overs at regular intervals to allow safe and flexible operation. Forming cross-overs in competent ground by tunnelling is relatively straightforward. However, in very soft ground forming a cross-over by bored tunnelling is difficult, expensive, and would require extensive soil improvement. Cross-overs were therefore placed adjacent to Newton, Toa Payoh, Raffles, Outram, Bugis and Marina Bay stations, where they could be formed using cut-and-cover methods as extensions of the main station excavation.

38. *Transition sections.* In the transition from tunnel to viaduct some relatively shallow tunnels were needed. These sections were outside the most densely populated areas, and therefore cut-and-cover tunnels were practicable. Cut-and-cover tunnels were used west of Tiong Bahru station, north of Toa Payoh station and east of Lavender station. Generally these excavations used strutted sheet piles for temporary support, except in a few locations close to building where the greater stiffness of diaphragm walls was preferred. Apart from their increased stiffness diaphragm walls had another advantage over sheet piles, in that they were not extracted. The extraction of sheet piles could cause settlement of buildings or the completed tunnels themselves.

39. *Out of town.* Although Singapore is relatively flat, the viaducts did intersect a few small hills. In two areas the route was carried through the hills by cut-and-cover tunnelling, to avoid energy inefficient changes of level.

Bored tunnelling

40. Approximately 11 route km was formed in twin bored tunnels. The MRTC's requirement was for a minimum clear internal diameter of 5·2 m with a stringent watertightness specification. A maximum overall leakage rate of $5\ \mathrm{ml\,m^{-2}\,h^{-1}}$ was required, with no 10 m length giving more than $10\ \mathrm{ml\,m^{-2}\,h^{-1}}$. This requirement was imposed to reduce consolidation settlements due to drainage into the tunnels.

41. All the tunnels were designed as well as constructed by the various contractors. Each contractor was able to choose the size of the access shafts, the tunnelling and lining method and the need for compressed air or soil improvement. Generally tunnels were driven from shafts located at the end of cut-and-cover stations or tunnels. The average length of drive between stations was 850 m. Only on the longest section between Toa Payoh and Novena stations, 1200 m, was a mid-tunnel shaft used.

42. Despite their relatively short length, most of the tunnel drives passed through a variety of ground conditions, from 150 MPa sandstones or granites, to stiff clays, to soft clays and loose sands. This can be seen from Fig. 4. Tunnelling methods therefore had to be flexible.

43. The most common method of tunnelling was to use a Greathead type shield equipped with rotary backhoe, backacter, boom cutter or roadheader (see Table 2). For soft ground sections the shields were typically provided with face rams and grids, and face stability was maintained by compressed air. Soil grouting was used either as a substitute for or in conjunction with the compressed air. All of the tunnels driven by the Greathead type of shield were lined using precast concrete segments.

44. Tunnel boring machines were proposed for the four drives of Contract 105,

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

and for two of the C301 drives. On C105 a single Hitachi-Zosen drum digger was proposed to complete all four drives between Somerset, Orchard and Newton stations in sequence, which required the completion of 3500 m of tunnelling in 18 months. This programme time included recovering and returning the shield to Orchard station three times and the concreting of all four tunnels. The drum digger erected a temporary lining of steel ribs and wooden laggings, with the permanent lining being cast in situ after each drive was completed. Compressed air was used for parts of the drives through weathered granite, and for the section through marine clay where the drum-digger was operated in a blind mode with the machine remoulding the clay before extruding it. Production rates did not meet programme requirements, and the fourth drive was tunnelled first by the new Austrian tunnelling method (NATM) then by a Greathead type of shield.

45. The drives between Lavender and Bugis stations on C301 encountered only Kallang formation and a little Old Alluvium, and thus provided ideal conditions for two earth pressure balance shields (EPBS). The shields erected precast concrete segments.

46. The NATM was used on four contracts. On C105 an initial drive in completely weathered granite used the NATM. Much longer NATM drives were undertaken on C104, predominantly in fresh or moderately weathered sedimentary rock, but with 250 m in completely weathered rock. For contracts 107A and 107B several short lengths of tunnel were required through the S3, and the NATM proved an excellent method in this formation. Even though overall rates of tunnelling were relatively, slow, the method minimized surface settlement. Fig. 15 shows the different temporary support details for C104 and C107A.

47. Compressed air was widely used in the tunnels driven by Greathead type shields and proved successful in maintaining stable conditions at the tunnel face even in the soft marine clay. However, there were several instances where air pressures could not be maintained due to permeable sands or fissured rocks. On occasions air blows occurred as far as 30 m ahead of the face. Soil grouting was then used to minimize the air leakage. Air capacity, including standby allowance, was generally 5–7 $m^3 min^{-1} m^{-2}$ and, except in two instances of major leakage, proved adequate.

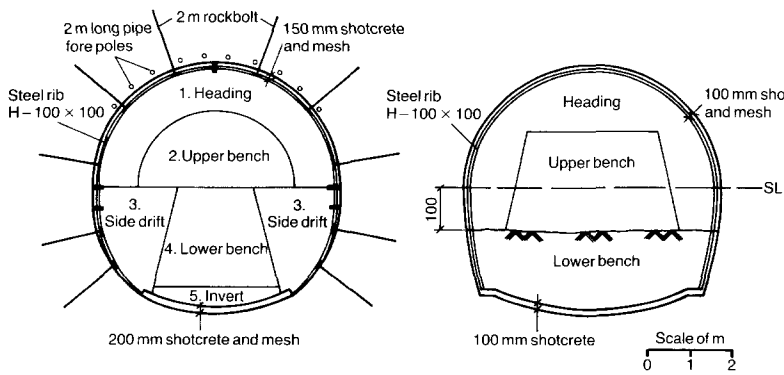


Fig. 15. NATM temporary works on (a) C104, decomposed granite face, 28 m overburden and (b) C107A, boulder clay face, 18 m overburden

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

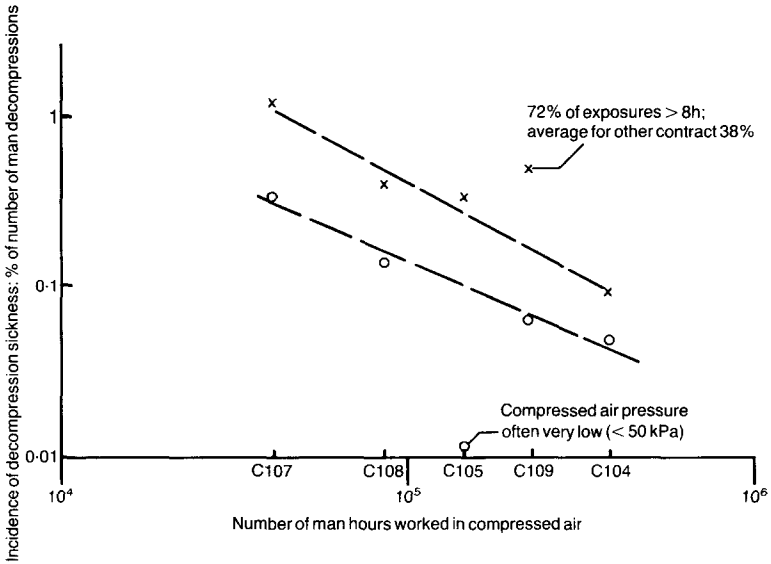


Fig. 16. Incidence of decompression sickness against length of exposure to compressed air

48. The health aspects of compressed air working were a major concern. Where possible contractors kept the air pressure below 100 kPa. A worker exposed to over 100 kPa was three times as likely to suffer decompression sickness as a worker exposed to less than 100 kPa. However, the rate of decompression sickness did decrease radically as the workers became acclimatized to it (Fig. 16). The Republic of Singapore Navy provided medical support and advice to the MTRC on compressed air working. Before being allowed to enter compressed air all workers were examined, and 26·5% were rejected as unsuitable. Man-lock and medical-lock attendants were trained at the Navy diving centre. Rates of decompression followed Singapore's Compressed Air Regulations based on the most recent data available.

49. Soil improvement was used extensively in the soft ground, either as a substitute for or in combination with compressed air. Applications included the following.

50. *Free-air tunnelling through soft ground.* Extensive use was made of jet grouting on contract 106 to allow the tunnels between Dhoby Ghaut and City Hall stations to be driven in free air through valleys infilled with soft clay and sands. On C107A a more limited length of one tunnel was treated with high strength silicate/reagent grouts to allow free-air tunnelling through corals and sands.

51. *Compressed air tunnelling through sands and fractured rock.* Where thick lenses of poorly graded sands were encountered compressed air proved to be less effective than in marine clay. Apart from the problem of air loss, it was also difficult to find a suitable air pressure. Too high a pressure completely dried out the top of the lens, causing it to run. Too low a pressure allowed the bottom of the

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

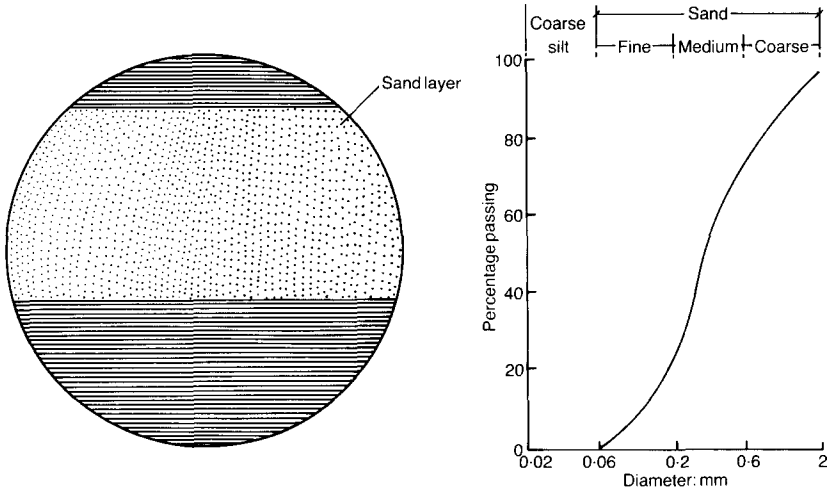


Fig. 17. Face on C104 needing chemical grouting

sand lens to flow into the tunnel. An example of the face condition and sand type where such problems were encountered on C104 is shown in Fig. 17. Similar cases on other contracts were treated with quick-setting, low strength silicate grouts to enhance the stability of the sand and reduce air loss. Fractured rocks were also treated in some areas with cement grouts for a similar purpose.

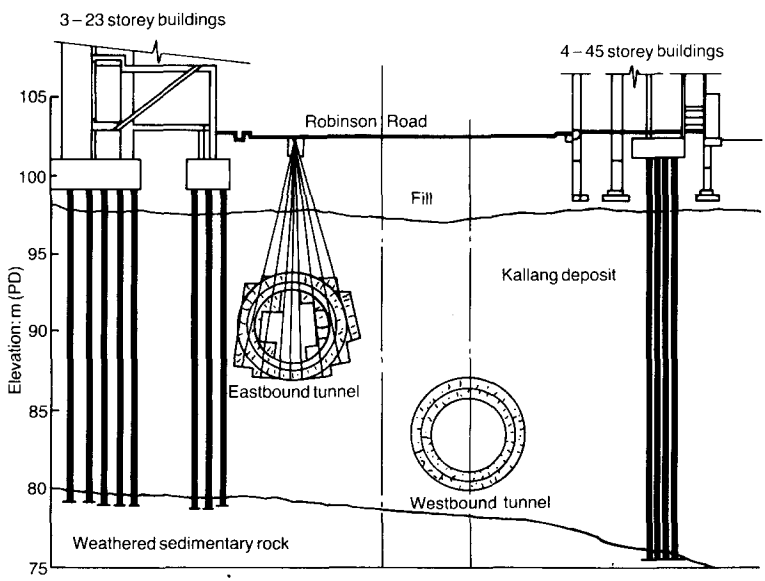


Fig. 18. Jet grouting at Robinson Road

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

52. *Reduction in building settlement.* Jet grouting was used to treat around 950 m of tunnel constructed under Robinson Road (C108) to reduce the settlement effects of tunnelling through marine clay on the buildings on either side (Fig. 18). This treatment required the formation of 4700 jet grout columns.

53. *Shield recovery.* In three instances of major inflows of soft ground into Greathead type shields, jet grouting was used to successfully recover the shields.

54. *Shield launching and docking.* Localized treatment using jet grouting was used as an aid to shield launching and docking in soft ground. The treatment for the end of the drives from Novena to Newton station is shown in Fig. 19.

55. *Sumps, cross-passages, back-shunts and other uses.* It was often inconvenient to provide compressed air for sumps in cross-passage or back-shunt construction. On some contracts soil improvement was used to allow this type of work to be done in free air. Various types of treatment were used, including jet grouting, chemical grouting and chemico-lime piling. Other uses of soil grouting included that shown in Fig. 19, pinning timber piles in place to allow them to be cut by an EPBS.

Settlements

56. As with any urban, underground project the effect of the work on nearby structures and utilities was a major concern, Singapore's buildings can be conveniently divided into three basic categories

(*a*) type 1: modern multi-storey commercial and residential buildings; either of reinforced concrete (RC) or steel frame construction, on substantial foundations

(*b*) type 2: government housing blocks; typically 6–12 storeys, of RC construction founded either on driven precast concrete piles or footings.

(*c*) type 3: shop houses; generally three storeys high, between 30 and 80 years old; of masonry construction, or more recently masonry infill to RC frame; typically founded on timber piles.

Settlement predictions were made both by the contractor and by geotechnical consultants. Dilapidation surveys were carried out on all buildings within the predicted zone of influence before the work started.

57. Singapore has always been a planned city, right from Sir Stamford Raffles' first concept, and this orderliness and the width of the major roads meant that it was possible to choose the alignment so that generally the tunnels did not have to pass directly under buildings. Those few buildings did, however, include some of considerable importance and age; among them the Sun Yat Sen Villa, the Chapel of the Holy Infant Jesus (CHIJ) and a Buddhist temple. The tunnels between Orchard and Somerset stations ran directly under the MRT's headquarters building.

58. Underpinning was only carried out as a precautionary measure at two buildings, a Government housing block at Braddell, underpinned by diaphragm walls (Fig. 20) and the CHIJ underpinned by mini- and micro-piles. The diaphragm walling was wholly successful, but the piling to the CHIJ itself caused up to 100 mm of settlement and considerable damage.

59. Soil improvement by jet grouting was used along Robinson Road and at the Sun Yat Sen Villa to limit settlement. In both cases the heave resulting from jet grouting caused some problems.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

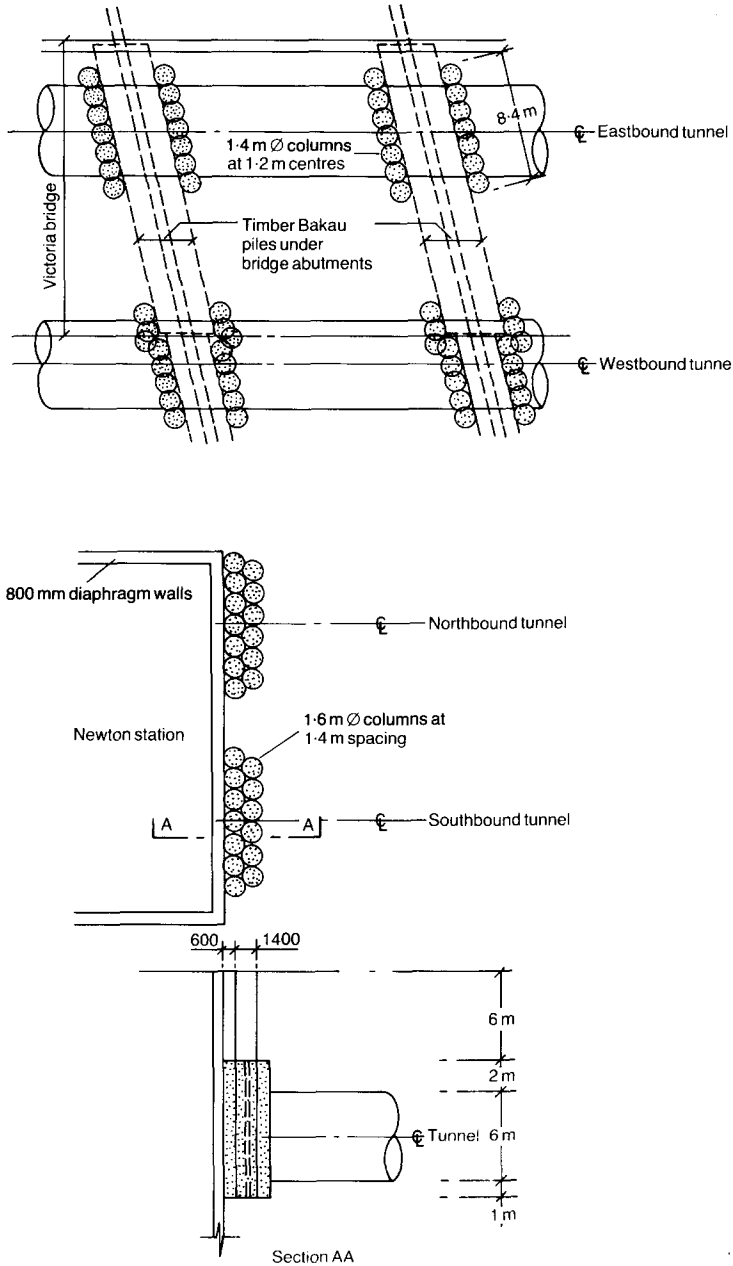


Fig. 19. Examples of jet grouting

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

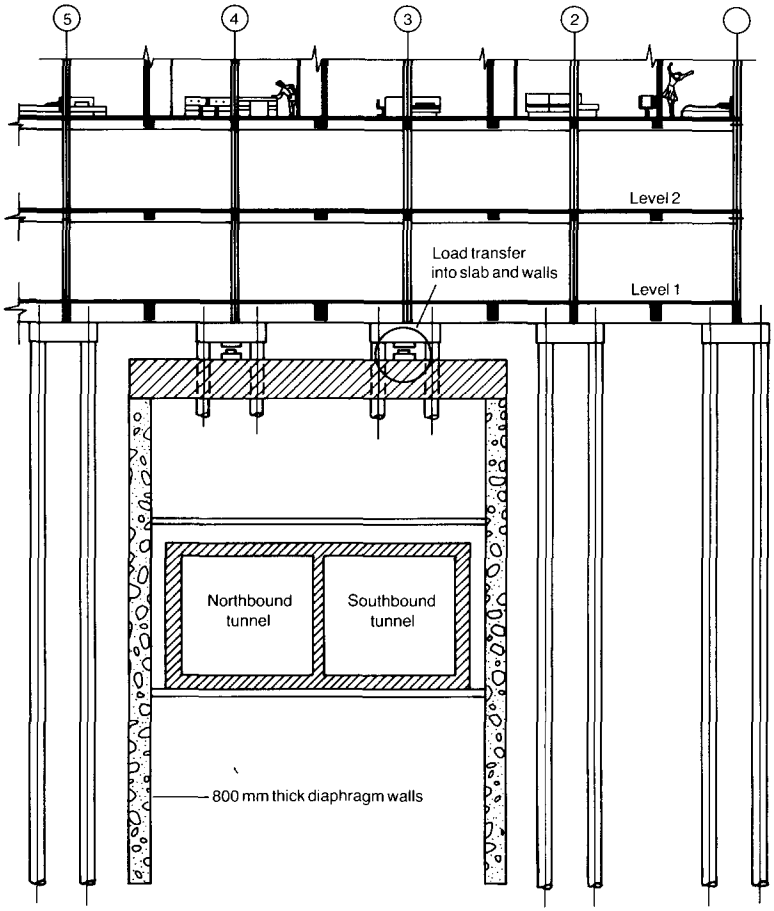


Fig. 20. Underpinning of block 206 at Braddell, showing completed tunnel box

60. Many thousands of surface and building settlement points were used to monitor the effects of construction. In addition use was made of inclinometers, piezometers and extensometers to separate the components of settlement. These components were

(*a*) installation: caused by ground movements due to the installation of the retaining walls (underground stations only)
(*b*) excavation: caused by ground movements during excavation of the tunnels or stations.
(*c*) consolidation: caused by pore-pressure changes generated during construction.

61. The installation settlements proved to be generally minor. There was some concern over movements during the installation of diaphragm walls through thick

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

*Table 3. Excavation rates by ground condition**

Contract	Method	S1/S2			G4/S4			Soft clays (marine and fluvial)			Mixed face			Bouldery S3, G3	
		Length: m	Rate: m/week		Length: m	Rate: m/week		Length: m	Rate: m/week		Length: m	Rate: m/week		Length: m	Rate average: m/week
			Avg.	Max.		Avg.	Max.		Avg.	Max.		Avg.	Max.		
104	Roadheader NATM†	850	17	36	—	—	—	—	—	—	—	—	—	—	—
	Compressed air NATM†	—	—	—	250	9	15	—	—	—	—	—	—	—	—
	Greathead, hand and backhoe	125	14	16	1490	22	35	1260	15	31	330	8	19	50	13
105	TBM†	—	—	—	1875	45	155	320	35	64	—	—	—	530	35
	Greathead with rotary backhoe	—	—	—	860	37	81	—	—	—	—	—	—	—	—
106	Greathead with rotary backhoe	—	—	—	1800	45	80	300‡	28	50	—	—	—	520	20
107	Greathead with backacter	—	—	—	2340	22	65	80	7	13	70	14	20	—	—
107A	NATM	—	—	—	—	—	—	—	—	—	—	—	—	130	9
108	Greathead with backacter	—	—	—	—	—	—	910‡	27	41	—	—	—	280	19
	Greathead with boom cutter	660	13	36	—	—	—	—	—	—	—	—	—	—	—
109	Greathead with roadheader	3560	24	50	—	—	—	—	—	—	—	—	—	—	—
301	Greathead with backacter	—	—	—	—	—	—	550	27	53	—	—	—	—	—
	EPBS	—	—	—	—	—	—	1595	40	63	—	—	—	—	—

* Average rates quoted are those from start to completion of a drive, but with the time for the following activities subtracted: air lock and shield installation, stopping the drive and grout ahead for sand lenses, concreting of in situ permanent linings (where used).
† Excavation and temporary lining rate.
‡ Jet grouted.

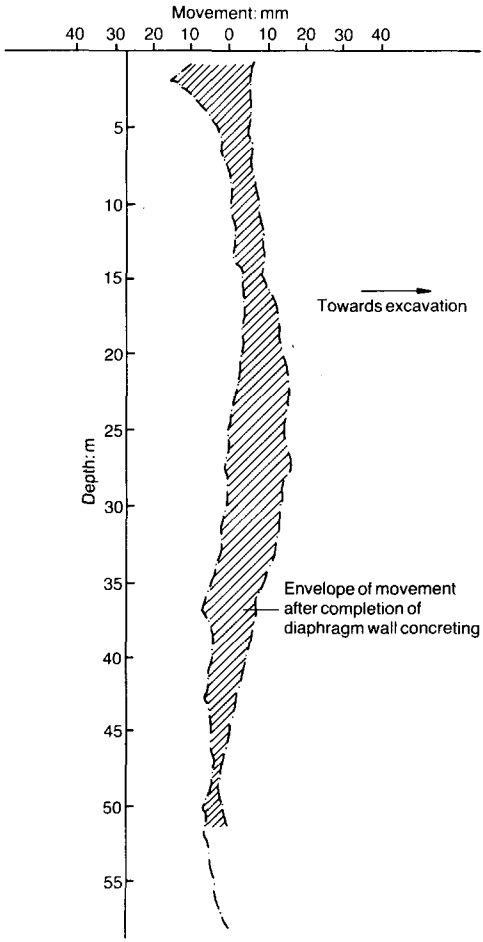


Fig. 21. Lateral movements due to diaphragm wall installation at Bugis station

deposits of the soft clays but these proved to be relatively minor. Fig. 21 shows the measured lateral movements caused by the diaphragm walling at Bugis station.

62. Settlements due to excavation at stations and cut-and-cover tunnels were found to be reasonably in agreement with predictions based on the excavated height (for stiff soils) and the lateral deflexion of the support system (for weak soils). Fig. 22 shows the deflexion and settlement at Newton and Bugis stations.

63. The immediate settlements due to excavation of bored tunnels were reasonably predictable in the soft soils, but predictions over the stiff soils were often seriously astray. The intermediate weathering grades of the Jurong formation were a particular problem, as shattered siltstone and mudstone often behaved more like a gravel, even though they were classified as rock. In one case nearly 400 mm of settlement occurred due to ground movements when the prediction was 20 mm.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

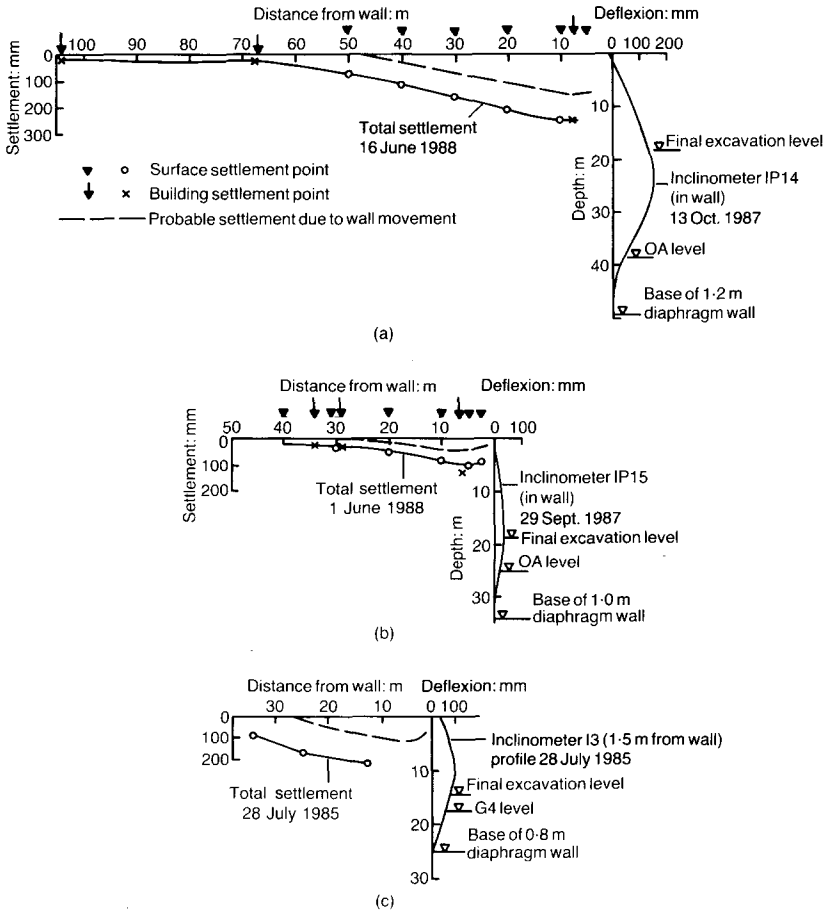


Fig. 22. Lateral deflexion and surface settlements at (a) Bugis station, (b) Bugis cross-over box and (c) Newton station

64. The settlement due to consolidation was more problematic, both for the stations and the tunnels. Both the lateral extent and the magnitude of these settlements were often underestimated. It can be seen in Fig. 22 that there was over 80 mm of consolidation settlement, and a trough extending over 100 m, at Bugis station despite diaphragm walls extending 30 m below excavation level and the use of recharge wells. Tunnels in fractured, permeable rock with overlying soft soils were the cause of some of the larger settlements recorded. Where the tunnel itself was in soft ground the measures adopted to control the face, compressed air or jet grouting, also helped to control the pore-pressure changes, although some settlement was inevitable whatever method was used.

65. The long-term consolidation settlements over tunnels driven through

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

marine clay have been studied in some detail. By plotting settlement against time on a logarithmic scale the consolidation can be identified, as it follows a straight line. Fig. 23 shows the settlement/log time plots for four different tunnels. The largest settlements, 200 mm over a single tunnel on C105, were clearly predominantly due to consolidation. This can be ascribed to the poor watertightness of the temporary lining made of steel ribs and timber lagging. The segmentally lined tunnels were far more watertight, particularly the C301 tunnels where water expansive seals were used. Average flows of 0·1 and 2·5 ml $m^{-2} h^{-1}$ were recorded on the two tunnels between Lavender and Bugis stations where the water expansive seals were used. Despite this high degree of watertightness a significant proportion of the total settlements recorded were still due to consolidation. There appeared to be at least three distinct causes of these consolidation settlements.

(*a*) On C104, the sand lens in the lower half of the face was probably dewatered by the tunnel, causing much of the consolidation.
(*b*) On C301A, the compressed air provided a support less than the K_0 pressure. This allowed the ground to move, generating reduced pore pressures near the tunnel and increased pressures further away.
(*c*) On C301B, the EPBS face pressure was significantly higher than the K_0 pressure, causing raised pore pressures around the tunnel.

There is a lack of piezometer data to confirm these hypotheses, but the varying lateral extent of the settlement troughs suggest different patterns of settlement.

66. Some examples of ranges of settlement experienced over the centre lines of the tunnels is given in Table 4. The table is very general, and covers many different reasons for settlement. However, two broad conclusions can be made.

(*a*) As discussed previously, settlements over tunnels in good ground were often larger than over tunnels in poor ground. The estimates of settlement made before tunnelling started tended to overestimate settlement in poor ground and sometimes very much underestimated settlement in good ground.
(*b*) It is quite clear that both method of tunnelling and the additional measures adopted (such as compressed air or grouting), could have a major effect on the degree of settlement in similar ground conditions.

Table 4. Ranges of final settlements over tunnel centre lines; last readings taken after a pair of tunnels had been driven

Ground type	Tunnelling method	Additional measures	Typical range of settlements: mm
S1/S2	Greathead shield	None	0–400
S3	Greathead shield	None	10–130
S3	NATM	None	10–60
G4	Greathead shield	Compressed air	10–20
G4	Drum digger	Free air	10–300
Kallang	Greathead shield	Compressed air	30–120
Kallang	Greathead shield	JGP and compressed air	30–60
Kallang	Greathead shield	JGP only	40–110
Kallang	Drum digger	Compressed air	200–400
Kallang	EPBS	None	70–140

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

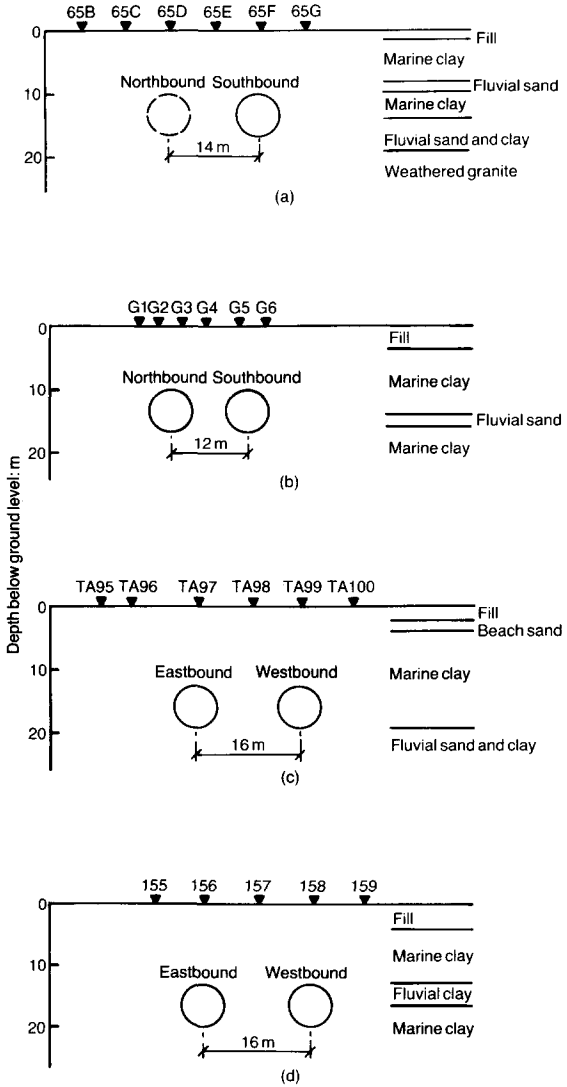
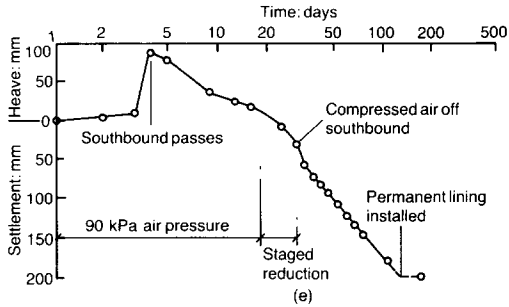
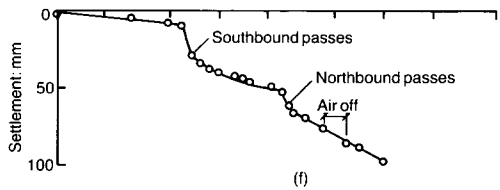


Fig. 23. Tunnel settlements against time on (a), (e) C105, drum digger, ribs and lagging; (b), (f) C104, Greathead shield, PCC rings, composite neoprene, bitumastic sealing strip, compressed air pressure 110 and 120 kPa in northbound and southbound tunnels, respectively; (c), (g) C301, Greathead shield, PCC rings, hydrotite gaskets, compressed air pressure 130 and 125 kPa in eastbound and westbound tunnels, respectively; (d), (h) earth pressure balance shield, PCC rings, hydrotite gaskets, face pressure 370/250 and 400/240 kpa in eastbound and westbound tunnels, respectively

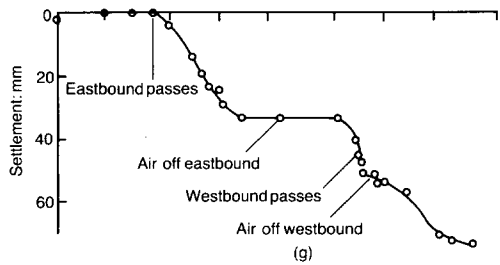
Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August



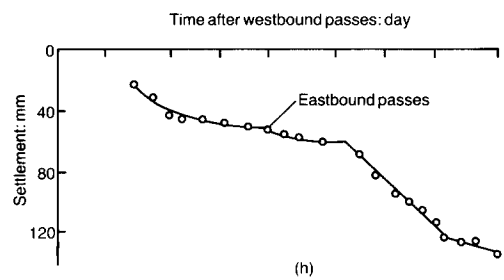
(e)



(f)



(g)



(h)

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Table 5. Jet grouting parameters

Contract	104/105A	108		301
Ground movement maximum: mm	+30	+550	+550	−10
Water pressure: MPa	30–35	—	1–2	−2
Grout pressure: MPa	0·5–2	20	20	20
Compressed air pressure: MPa	0·7	0·6–0·7	0·6–0·7	0·7
Water flow rate: l/min	65	—	30–40	30–40
Grout flow rate: l/min	65	60	60	60
Air flow rate: m^3/min	1–3	1–3	1–3	2–3
Rotation speed: rev/min	5–15	5–10	5–10	5–15
Withdrawal rate: min/m	20	11–22	11–16	16
Casing: mm	135	—	—	250
Additional comments		Both methods used concurrently		Downward stages of 2m

It was also noted that the high degree of watertightness of the C301 tunnels appeared to cause a delay in the onset of the consolidation settlement. This delay was of about a month after the driving of the second tunnel (301B) or the decompression of the second tunnel (301A). As a result the maximum rate of settlement often occurred just as the frequency of monitoring was being reduced.

67. It is worth noting that the largest ground movement recorded during MRT works was during neither tunnelling nor cut-and-cover construction, but during preliminary soil grouting for the tunnels. Large heaves, of up to 1·6 m but more typically 200–400 mm, were recorded during large scale jet grouting of marine clay. Nozzle pressures of 10–20 MPa were typically used for jet grouting. The pressurized jet both cut the soil and lifted the cuttings up an annulus around the rods. The low passive resistance of soft clay meant that any blockage of the annulus quickly translated into outward movement of the soil and consequent heave. Methods to eliminate the potential for ground movement by grouting in downward stages, or by using casing and precutting, were developed for use in sensitive locations and proved effective. Examples of the jet grouting parameters and the maximum surface movements are shown in Table 5. Some heave, up to 100 mm, was also recorded over areas of chemical grouting where the gel time was short (15–60 s). Within the context of the settlements recorded in Table 4 even this degree of movement is significant.

Building and utility damage

68. Generally the settlements due to the excavation of tunnels or stations caused less building and utility damage than the settlements due to consolidation or the heaves associated with jet grouting. The settlements due to consolidation tended to be widespread and have a relatively flat profile. As such they did not produce the angular distortions which can cause major structural damage. So they did not pose a threat to the public, but had considerable nuisance value. Damage included

(*a*) Separation of utilities and external apron slabs from buildings founded on piles.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

(*b*) Separation of ground floor slabs from piled ground beams. This typically occurred where non-suspended floors had been placed directly onto fill overlying organic clays and it was often difficult to determine whether any damage was related to MRT works or whether the situation had been developing for years.

(*c*) In one case only, the failure of bored piles due to the negative skin friction induced by the settlement.

69. Heave due to jet grouting caused minor damage to the outhouse of the Sun Yat Sen Villa, and, elsewhere, several hundred thousand pounds worth of damage to utilities and road carriageways. Settlement due to mini- and micro-piling at the Chapel of the Holy Infant Jesus caused structural distress.

Depots

Site formation

70. The three depots, at Bishan, Ulu Pandan aad Changi occupy 35, 9·0 and 16·6 ha, respectively. Finding these large, clear areas in a densely populated island was not easy, and the areas which were available were inevitably those unattractive to others. As a result poor subsoil conditions were encountered at all three depot sites.

71. At Bishan Depot a layer of soft (water content up to 60%), organic estuarine clay 5–7 m thick covered much of the site. Sands and clays of the Kallang formation underlay the estuarine clay, giving a total depth to the underlying weathered granite of 7–25 m.

72. The whole site had to be raised by 2·2 m to avoid flooding. If this had been done simply by filling, the combination of load from the fill and the ballasted tracks would have caused an estimated 2·5 m of settlement over 30 years. It was therefore decided to carry out trials to see if a suitable method of ground improvement could be found.

73. The methods selected for trial were: preloading with vertical drains; dynamic replacement and mixing (DRM) and preloading; DRM combined with vertical drains and preloading. The details and costs of the trials are shown in Table 6. The DRM, which used a heavy weight to drive sand into the soft clay, proved an effective method of both accelerating and limiting the settlement due to preloading. However, because of the short programme time available it was finally

Table 6. Costs of test embankments at the Bishan depot

Embankment	Area: m^2	Cost: S$/$m^2$*	Height: m	Ground improvement method
A	1600	61	3·6	Preloading and drains 2·0 and 1·5 m spacing
B	1800	124 202†	3·0	DRM and drains at 2·2 m spacing
C	1600	36	3·5	Preloading only (no drains)
XI	1500	119 197†	3·9	DRM only (no drains)

* Excluding costs of instrumentation, site clearance, offices and insurance.

† Extrapolated cost for DRM with heavy tamping.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

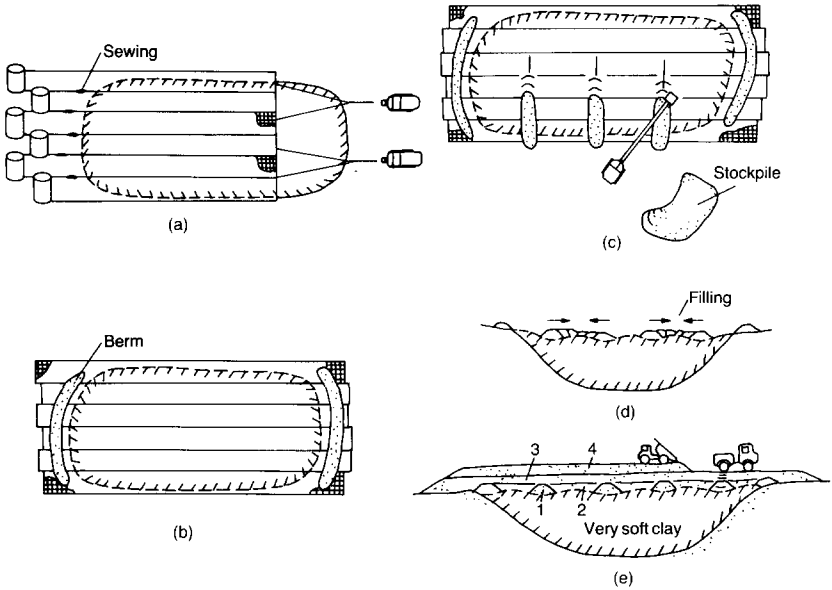


Fig. 24. Method of filling over Changi depot ponds: (a) stage 1: placing fabric; (b) stage 2: stabilizing berms; (c) stage 3: construction of berms; (d) stage 4: widen berms; (e) stage 5: layered filling

decided to use the only marginally more expensive solution of a concrete deck supported on precast concrete or steel piles. 15 000 piles were driven starting in November 1984, with track laying starting exactly a year later. The piles supported 120 000 m^2 of deck at an average cost of S$250/$m^2$ of piled deck.

74. At Ulu Pandan the site consisted of hillsides of weathered Jurong formation around a valley. Up to 6 m of soft sediments had accumulated in the base of the valley. Owing to the shallow depth of soft deposits, it was economical simply to remove the soft soil and fill it with material excavated from the hillsides. The extensive earthworks required cost S$57/$m^2$ of useable area.

75. At Changi depot there was up to 20 m of Kallang formation, mostly marine clay and fluvial deposits, underlying the site. Although compressible, these were not as bad as the estuarine clay which had caused the major problem at Bishan depot. The programme for Changi was less critical than it had been for the main depot at Bishan, and it was decided that vertical drains and preloading would be a suitable method of treatment. This required filling to an average height of about 6 m above the existing ground level. One problem, however, was the presence of large ponds covering about 4 ha of the site. These ponds were the results of sand quarrying. The tailings from the extraction plant had been replaced in the ponds, resulting in up to 8 m depth of exceptionally soft material. A shear strength of 1 kPa was measured 2 m below the surface of the tailings. Similar ponds in the vicinity had been treated by extraction and replacement, but for the depot it was considered adequate to leave the tailings in situ and treat them by preloading and vertical drains. To apply the preload to the very soft tailings a

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

detailed construction sequence using high strength geotextile and a careful fill placement was specified (Fig. 24). The cost of the earthworks at Changi Depot was S$99·70/m^2.

Viaducts

Piling

76. Over two thirds of the MRT route, about 47 km, is on viaduct. Good quality rock was close enough to the ground surface to allow the viaducts to be supported on footings at very few locations. The majority of the piers are therefore supported on piles taken into weathered rock or Old Alluvium. The piles commonly had to penetrate varying depths of soft Kallang deposits before reaching the bearing stratum. These soft deposits could be deep, and more particularly so under phase 2A.

77. In general, cast in place reinforced concrete bored piles, varying from 600 mm to 1500 mm dia. but usually 1000 mm dia., were installed to support the foundations of the viaduct piers. In addition to bored piling, 325 mm square reinforced concrete driven piles and 600 mm dia. steel tube raking piles were also used on the phase 2B contracts.

78. Crane mounted drilling rigs equipped with rotary augers were the standard plant used for the bored piling. The drilling rigs were assisted by service cranes for installing casings, reinforcement cages and the tremie pipes which were used to place the 30 N/mm^2 ordinary Portland cement pile concrete. The piles were drilled to average depth of 20–25 m along the north and west lines. However, bored piles of up to 70 m deep were required in places along the east line where very soft soils exceeding 40 m deep were encountered. Temporary casings were necessary in soft unstable soils and permanent casings were sometimes installed in the very soft marine clays. There were difficulties in driving some of the casings, due to the presence of boulders, problems of withdrawal and delays due to welding extensions to casings for the deep piles. As a result, some contractors subsequently used bentonite to support the borings to avoid the problems encountered using casings.

79. Before starting the main piling on a contract, preliminary piles, in some cases instrumented with strain gauges and extensometers, were tested to 2·5 times working load to check the efficiency of the design. During the piling routine testing was carried out as shown in Table 7. These included tests to 1·5 times working load, with the reaction provided by kentledge (most commonly), anchors or tension piles. Extensive non-destructive testing was also carried out, particularly for the bored piles. Some details of the cost of these tests and their application based on MRT experience is given in Table 8. Generally the piles tested proved satisfactory, although some defects were picked up by the testing. These included: soft toe (dynamic pile testing and base coring), cracks and cold joints (dynamic pile testing, modified shock and vibration), low driving energy (pile driving analyser) and segregated concrete at the pile toe (base coring). The most common remedial method for such defective piles was to provide additional piles to compensate. In some instances base grouting was used to improve piles with a poor toe.

Pile caps

80. Pile caps were reinforced with top and bottom mats interlinked with pile-head rebar and column starter bars. Pile-cap sizes have varied according to each

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Table 7. Routine pile testing, types and percentages

Test type		Bored pile	Precast concrete pile*	Steel H pile*	Steel pipe pile	Caisson
Total number of piles		21 210†	22 500	2914	1618	6
Preliminary pile tests		80	42	10	4	
Working pile tests		254	233	10	6	
Lateral pile tests		39	—	—	—	
Dynamic pile tests		45	82	—	8	
Vibration tests		656				
Modified shock tests		418				
Seismic shock tests		38				
Sonic logging		233				
Base coring		52				
Piles tested: %	Maintained load test and lateral load test	1·4	1·0	0·3	0·4	—
	Non-destructive test	6·8	0·4	—	0·5	—

* Includes 15 000 piles for Bishan depot.

† Includes 1600 piles for Marina Bay station and tunnels drilled by RCD.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Table 8. Routine pile tests, cost and applications

	Small strain tests			Large strain tests		Other tests	
	Vibration tests	Modified shock test	Seismic test	Maintained load test	Stress wave measurement	Sonic logging	Base coring
Cost: S$	300	280	230	(6–25)/t	500–2000	300	600
Depth limitation	$<30D$	$<30D$	$<30D$	Unlimited	Depends on energy input and concrete stress	Unlimited but difficulties arise from joining of the sonic tubes	Unlimited but difficulties arise from joining of the coring tubes
Maximum depth for MRT: m	—	—	—	65·0	60·1	65·0	65·0
Bearing capacity	Very little although dynamic stiffness can sometimes be used as a comparison	Same as vibration test	NA	Excellent—but may give a false picture in thick deposits of soft clay which may in future cause down drag on the pile	A good indication if calibrated against maintained load test	NA	An indication of end bearing capacity
Structural integrity	Fairly good	Fairly good	Fairly good	Indirect	Good	Good	Good (base only)
Speed	Fast	Fast	Fast	Slow	Fast	Fast	Slow

Table 9. Pile cap details

Phase	Pile cap size, length × width × depth: m	Numbers of piles*	Concrete grade: N/mm^2	Comments
1	8·0 × 8·0 × 2·5	9B	30	Single track
1A	4·8 × 4·8 × 1·5	4B	30	Single track
	7·8 × 4·8 × 1·5	6B	30	Twin track
	7·8 × 7·0 × 1·5	8B	30	Twin track
2A	7·8 × 4·8 × 2	6B	30	Single track
	7·8 × 7·0 × 2	8B	30	Twin track
2B	7 × 5 × 1·75	5 or 6B	40	Single track
	8 × 7 × 1·75	8B	40	Twin track
	8 × 6 × 1·75	26D	40	Twin track
	8 × 6 × 1·75	14D +12DTS	40	Single track

* B = 1000 mm dia. bored RC piles; D = driven 325 × 325 mm precast RC piles; DTS = driven 600 mm dia. tubular steel piles.

consultant's design and typical details are given in Table 9. In addition to those shown, some pile caps had to be enlarged to accommodate increased loadings resulting from greater crosshead sizes, column heights or track curvature. Space constraints due to existing utilities services also often affected pile-cap dimensions.

81. Pile caps adjacent to roads, existing structures, soft ground or in deep cut were excavated between driven steel sheet piling with structural steelwork walers. Where good ground occurred the sides were battered to a stable angle of repose. In general pile-cap construction did not cause major problems or programme constrains as acceleration of this activity could be easily achieved with an increase of resources and ample work areas were usually readily available. However, in some places, particularly the large pile caps of the in situ bridges, construction was often slow and the contractors had to ensure that they started this work as early as possible to avoid potential delays.

Columns

82. Apart from a few exceptions, the viaduct piers were either 1750 mm dia. circular columns for single track supports or 1750 mm by 2150 mm elliptical columns for twin track supports. The 40 N/mm^2 concrete columns were reinforced with two concentric rings of Y40 and Y50 main bars tied with a system of Y12 or Y10 closed links usually at 150 mm centres.

83. To achieve a high quality finish, comparable with the precast beams, steel formwork was used for column construction. The columns have vertical feature grooves 50 mm wide by 15 mm deep at 600 mm centres. These grooves enabled the column formwork to be assembled in four sections with all vertical construction joints hidden in the feature grooves so that a high quality concrete finish was obtained. Horizontal construction joints were minimized or eliminated by using continuous moulds, at least 6 m high, for standard height columns. A 150 mm dia. cast iron downpipe is cast into the centre of each column to carry viaduct storm water to the ground level drainage systems. Columns were concreted in one pour, usually by concrete pump. The contractors had to procure enough sets of moulds to ensure that column construction would proceed ahead of and not delay the

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

critical crosshead construction. The average column cycle time was 5–6 days from the start of reinforcement fixing to concreting.

Crosshead construction

84. Most contractors opted to use steel moulds for the in situ reinforced concrete crosshead construction to ensure a high standard of finish and dimensional accuracy. The high density of rebar in the crossheads initially resulted in slow cycle times until steel fixers became familiar with the work. Production rates then usually settled down to average 8–12 days for single track crossheads and 10–15 days for twin track. However, these periods could vary substantially depending on the size, location and complexity of individual crossheads (Fig. 25).

85. Typical beams were based on standard spans of 25 m between pier centres; however, by varying the lengths of the in situ concrete crossheads, spans of up to 30·5 m were attained. This had the advantage of providing a constant depth of crosshead/beam profile and minimized the types of beams to be constructed. This policy was not fully adopted by the phase 2A designers where longer and deeper precast beams were used at some locations to span ground-level constraints such as road junctions. The crossheads had also to be deepened to suit.

86. The profile and cross-section of the crossheads are identical to the viaduct beams and the adoption of half joints at the bearings supports enables the beam profile to be carried through without disruption. Very accurate setting-out of the crossheads horizontal and vertical alignments and, in particular, the half joints was necessary to ensure no subsequent problems would occur when placing the precast beams. The 40 mm nominal gap at the crosshead and beam joints allowed little margin for error and it was a requirement that each beam at the precasting

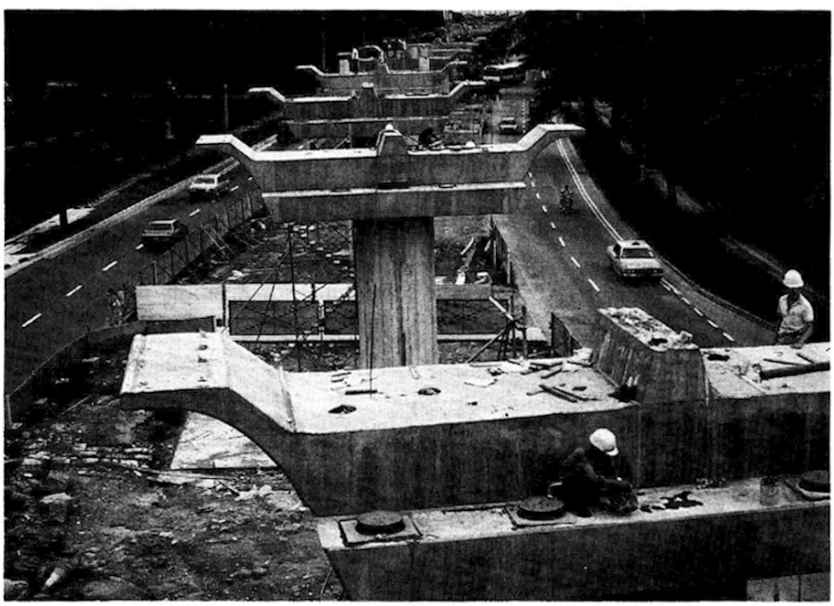


Fig. 25. Twin crosshead construction and bearings preparation

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

yard had to be individually checked against its as-constructed crossheads before concreting to ensure an accurate fit.

87. Although crosshead construction was relatively straightforward, it was an important critical path operation and some contractors initially underestimated the time and resources necessary for this work. Delays to crosshead construction would result in knock-on delays to precast beam manufacture and launching so that additional crosshead forms, support steelwork/scaffolding and skilled labour had to be mobilized as quickly as possible to maintain programme.

Bearings

88. The beams were supported on the crossheads either by laminated rubber bearings as on phase 1, C102 contract or by mechanical pot bearings on all other contracts. The laminated rubber bearings were combined with shear pins, one at each end of the beam, to resist longitudinal and transverse loads.

89. For phase 1A the designers used four pot bearings per beam which were sandwiched between top and bottom locating plates to aid future replacement of the bearings. After installing the bottom locating plates, the pot bearings together with their top locating plates were placed in position and the viaduct beam lowered. The top locating plates were bonded to the beam using epoxy resin which was poured down a tube cast in the beam at each bearing location. Similarly, four other tubes located at each corner of the bearing plates acted as vents to ensure complete contact of the epoxy between top plate and beam soffit. The beam was held in position by the launching gantry for 4–6 hours until the epoxy cured and the load could then be transferred.

90. Phase 2A and 2B designers used three pot bearings per beam, a single fixed bearing at one end and two partially guided bearings at the other end. The method of installation was similar to the phase 1A bearings but no top and bottom locating plates were used. The bottom base plate of the bearing was grouted into position using non-shrink cementitious grout and the top bearing plate was coated with a thin layer of grout to act as a bond breaker. After the beam was lowered, it was supported by the launching gantry or by temporary jacks until the poured epoxy had set. The bearings have been detailed so that they can be replaced in their exact position by using top and bottom locating bolts which can be unscrewed.

Waterproofing membrane

91. For the phase 1/1A viaduct the consultants specified a bituminous based waterproofing membrane with a woven plastic braid which was bonded to the viaduct concrete. This membrane was easy to lay and the site staff had few problems controlling the workmanship. The phase 2A and 2B consultants specified an ethylene-copolymer-bitumen plastic membrane with thermoplastic properties. The membrane was not bonded to the viaduct concrete and, being a stiff material, was often difficult to install on the sloping and angled surfaces. It also had a tendency to corrugate under strong sunlight.

Cable supports

92. Two types of precast concrete panel-type units were installed on the viaduct to provide a support for the power, signals and communications cables and also to contain the ballast. These units were usually cast off site and were placed in position using a lorry-mounted hydraulic lifting boom running on the

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

screeded viaduct. Holes and sockets cast in the units allowed cable brackets and sunshields to be fixed. The units were approximately 700 mm high with wall thicknesses of 80 and 100 mm.

Viaduct fascia and handrails

93. Along the outside face of the phase 1/1A and 2B viaduct walkways an aluminium handrail and fascia were installed. The handrail was formed of twin horizontal rails supported by 1 m high posts at 2·5 m centres. A curved aluminium fascia 300 mm deep was suspended from the base plates of the handrail posts to provide a smooth line along the cantilever edge. Attached to the fascia at the piers were lightning conductor tapes. For the phase 2A contracts the consultants changed the fascia design to a precast concrete unit surmounted by a tubular aluminium handrail and posts which were bolted onto the concrete fascia.

Movement joints

94. Buried movement joints having average widths of 40 mm were detailed between the precast beams and the crossheads. The waterproofing membrane was carried over the joints and protected by tiles or a stainless steel plate. Surface movement joints with neoprene insets and protected by stainless steel plates were provided at the larger spans of the in situ bridges.

Precasting yards

95. For the elevated sections of the MRT phases 1 and 2, 13 contracts were awarded and 16 casting yards were established to produce the beams. A total of 3341 precast viaduct beams were manufactured in just under three years. The mass production of these very large beams of average weight 170 t and 22·3 m span demanded a high degree of automation because their production rates would have the greatest single influence on the tight viaduct construction programmes. Beam production peaked from November 1986 to February 1987 when the total casting yard production averaged 260 beams per month.

96. The method of construction envisaged for the viaduct contracts required that the beams be precast in yards adjacent to the railway alignment. The MRTC provided the sites for the precasting yards and it was recognized that the length of viaduct that could be constructed in the contract period would be largely dependent on the casting yard location and size. A casting yard having a small area with little storage space would severely restrict beam production rates and limit the speed of viaduct construction. Conversely, a large casting yard would allow the provision of extra beam moulds and increased storage facilities which would enable flexibility to be built into the viaduct construction programming. After manufacture, the beams would be lifted from the precasting yards onto the previously constructed part of the viaduct where they would be transported on a temporary rail track to a self-travelling launching gantry which would place them in their final position on the viaduct crossheads (Fig. 26).

97. All contractors used steel moulds for the manufacture of the precast beams and they had to carefully evaluate the periods required to fabricate, deliver and set-up the moulds in the casting yards. The other major items of casting yard equipment were the specialist gantries used to service the construction of the beams and for lifting the beams from the casting yard onto the viaduct. The yard gantries were either separate gantries, one for beam manufacture and another for

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

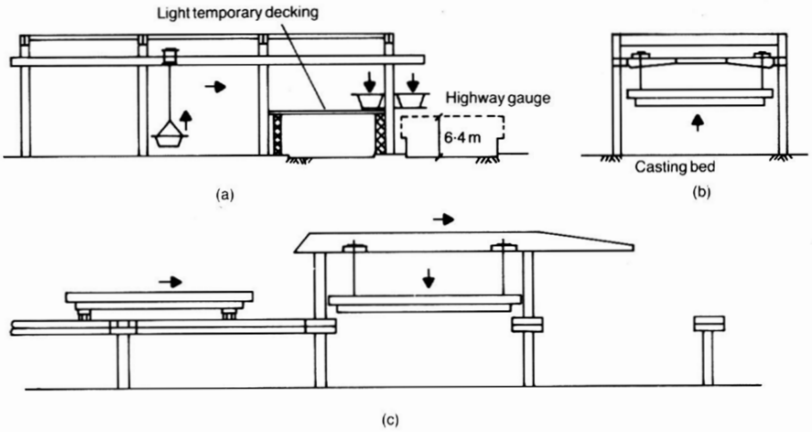


Fig. 26. Viaduct construction method: (a) side access from casting yard onto median in road; (b) end view; (c) erection of precast box girders using a launching girder

transporting from yard to adjacent viaduct, or a single gantry which served the casting yard and loaded the beams onto the viaduct. The yard gantries travelled on parallel rails and straddled the precasting beds. The gantries performed a number of functions including lifting and placement of mould forms, rebar cages and concreting equipment. They also removed completed beams from the moulds and transported them to storage areas (Fig. 27).



Fig. 27. Viaduct casting yard and gantries

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

98. Wherever possible casting yard layouts were designed so that the longitudinal axes of the beams were parallel to the line of the viaduct. This allowed relatively simple transition of the beams from the beds onto the adjacent viaduct. However, at two yards, area constraints caused the beds to be built with the longitudinal axes of the beams at 90° to the line of the viaduct. As a result of this, turntables had to be provided to move the yard gantry through 90° to place the beams in their correct alignment on the viaduct. For four phase 2A contracts the design required some longer span heavy beams to be manufactured, so both the yard and launching gantries had to be capable of lifting beams of up to 270 t. These heavy beams considerably increased the cost of the gantry equipment when compared to the gantries used for the phase 1A and 2B contracts where the maximum weight of beams was 170 t.

Beam casting techniques

99. A typical viaduct precast beam 22·3 m long weighing 170 t contains 16 t of rebar, 2·35 t of prestressing strand and 67 m^3 of 45 N/mm^2 grade concrete. The typical beams were post-tensioned using four tendons of multiple seven wire super strand complying with BS 5896: 1980. Tendons were single end tensioned from alternate ends of the beams.

100. Five casting methods were used by MRT contractors for manufacture of the viaduct beams. The methods are described in Table 10 and relevant statistical data are given in Table 11.

101. The majority of contractors used either method 2 or 3 for casting the beams. The two methods are similar except for the concrete pours which were either single or two stage concreting, respectively. Both methods used short lengths of bolted internal steel formwork to form the hollow box section. The labour forces and cycle times for both methods were very similar with method 2 having a possible slight advantage on cycle times but this was very dependent on individual yard efficiency and management.

102. The cast in situ construction adopted in method 1 was almost identical to method 3 and utilized the same technique of two stage concreting. The much slower cycle times were a result of the multiple falsework erection and dismantling requirement together with the problems of achieving efficient management of labour and resources when beams are not made in the factory-like environment of the casting yards.

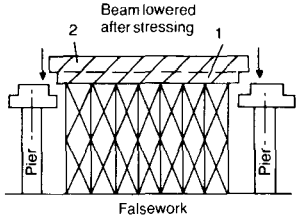
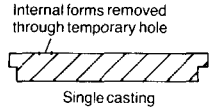
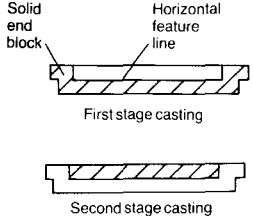
103. Method 4 also utilized a two stage concreting sequence. However, the contractor was able to reduce his cycle times by utilizing a more sophisticated set of internal forms. These forms allowed half the total length of the beam to be cast complete, together with the bottom slab and half height webs of the remaining half of the beam. The internal forms were easily removed via the open central part after the first casting and the relatively short length of top slab and upper webs remaining could then be quickly formed and concreted. This achieved optimum cycle times at least three days less than those achieved for methods 2 and 3.

104. Method 5 differs from the other methods in that it is a segmental form of precasting. The complete cross-section of the beam is cast as a central segment and the internal moulds are removed from each end on fixed rails. The central segment is then lifted from its casting bed and taken to the end block casting bed by the overhead gantry. The two end blocks are then cast onto the central segment. With the achievement of a seven-day cycle time this was the quickest method of con-

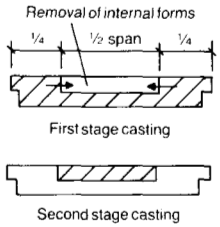
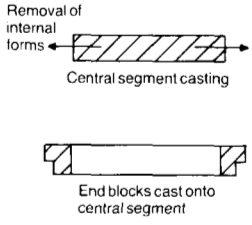
Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Table 10. Beam casting methods

Method	Illustration	Sequence of construction	Comments
1. Cast in situ using two stage concreting		1. Falsework erected between viaduct piers. 2. Steel moulds assembled on falsework and rebar/prestress tendons installed. 3. Lower webs internal forms fixed. 4. First stage concreting of bottom slab, end blocks and half height webs. 5. First stage internal forms removed and second stage internal top slab/webs fixed. 6. Top slab rebar and side forms fixed. 7. Second stage top slab/webs concreting. 8. Internal forms removed and beam post-tensioned. 9. Remove beam moulds and dismantle falsework. 10. Lower beam onto viaduct crosshead bearings.	This method was used by one contractor. The main advantage was that no casting yard had to be set up nor was a launching gantry necessary to place the beams. Disadvantages were the much slower cycle times for beam casting (optimum 16 days average 24 days) and the substantial numbers of moulds and associated resources required to meet the construction programme.
2. Casting yard single concrete pour		1. Prefabricated rebar cages placed in moulds 2. Prestress tendons installed. 3. Internal forms fixed. 4. Top slab rebar fixed. 5. Beam concreted in single pour. 6. Internal forms removed though top hole which is later concreted. 7. Beam post-tensioned and removed.	Four contractors used this method of beam casting. Optimum cycle times varied between 10 and 12 days after initial learning period. A uniform concrete finish is achieved.
3. Casting yard two stage concreting		1. Prefabricated rebar cages placed in moulds for bottom slab/web and end blocks. 2. Prestress tendons installed. 3. First stage concreting of bottom slab, lower webs and solid end blocks. 4. First stage internal forms removed and second stage top slab internal forms fixed. 5. Top slab rebar and top side forms fixed 6. Second stage top slab/webs concreting. 7. Internal forms removed, side forms released. 8. Beam post-tensioned and removed.	Six contractors used this method of beam casting. Optimum cycle times varied between 11 and 13 days after initial learning period. Care is required for colour compatibility of two stage concreting and construction joint detail.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/licep_1989_2628.pdf by National University of Singapore user on 18 August

4. Casting yard two stage concreting		1. Prefabricated rebar cages placed in moulds for both ends quarter spans, bottom slab and lower web half span. 2. Prestress tendons installed. 3. First stage concreting. 4. Remove internal end moulds and first stage lower web forms. 5. Install centre half span internal top forms and fix top slab rebar. 6. Second stage concreting. 7. Post-tension and remove beam.	One contractor used this method which allowed the complete cross-section of the end quarter spans to be concreted together with the centre half span bottom slab and lower webs. The internal forms of the end quarter spans were easily removed via the centre half span. Improved cycle times of eight days were achieved by this casting method.
5. Casting yard two stage segmental casting		1. Prefabricated rebar cages are placed in the central segment moulds and internal forms fixed. 2. End forms closed and central segment cast. 3. Central segment lifted and placed at end blocks casting bed. 4. End blocks rebar and internal forms fixed. 5. End blocks cast onto central segment. 6. Prestress tendons installed and tensioned. 7. Beam removed to storage area for tendon grouting and finishing.	One contractor used this segmental method of construction. The central segment moulds casting cycle times are three days and the end blocks 4–5 days. A complete beam can be made in seven days. Care is required for colour compatibility between the central segments and end blocks castings. Joint details are also a problem at the segmental casts.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Table 11. Beam construction data

Method	MRT contract	Mould sets	Beam production		Construction period: month	Total number of beams	Prestress system
			Planned cycle time: d	Monthly planned productivity			
1	C102	10 (min.) 32 (max.)	24	15	13	173	VSL
2	C202	6	12·5	14	15	202	PSC
	C403	10	11	24	12	289	BBR
	C304	10	14	15	14	208	BBR
	C306 Y1	8	12	20	10	202	BBR
	Y2	8	14	17	14	238	BBR
3	C203	10	12	24	13	256	VSL
	C204	6	11	14	11	146	VSL
	C303	7	14	13	13	186	VSL
	C305	9	13	18	12	186	VSL
	C404	11	11·5	20	11	226	VSL
	C405 Y1	6	11	14	12	133	VSL
	Y2	8	12	16	18	206	VSL
	Y3	7	12	16	7	94	VSL
4	C302	7	8	25	13	210	PSC
5	C402	10	8	32	14	386	PSC

struction of the five methods but had the disadvantage that it was not possible to conceal the segmental jointing and careful concrete quality control was essential to ensure reasonable colour compatibility of the segments.

105. Although five casting methods were used the items of work for the construction of all the beams were similar. As an example, the detailed sequence of construction for method 3, used in phase 1A contract 203, will now be given.

Precasting method 3

106. The beam reinforcement was assembled in jig beds located in the casting yards and the rebar cages were then placed into the steel moulds which had previously been cleaned, oiled and all dimensions checked. The prestressing ducts were placed in position and strand installed either by hand or with a pushing machine. The ducts were secured by profiled rebar supports, both top and bottom to prevent displacement during concreting. The internal forms to the bottom half of the webs were fixed and, in the solid end blocks of the beams, lifting hooks, made out of prestressing strand, were installed together with the tubes for grouting of bearings.

107. An important part of the checking was to ensure that curvature, due to the longitudinal alignment of the railway, was accurately set out along the top cantilever edges of each beam. The beam curvature had to match the as-built curvature of the crossheads onto which the beam would be placed. Finally the end forms at the tendon anchorages were closed and checked before concreting the first lift, usually three days after installing the rebar cage in the mould.

108. The web internal moulds were removed the following day after concreting and the construction joints prepared using a high pressure water jet. Internal forms for the top slab and upper web of the box girder were installed by day 5. Top slab rebar and joint construction separator plates in the flying cantilevers were then fixed, together with cast-in items. For stray current corrosion protection the cast-in items had to be at least 25 mm away from all rebar. Welded mesh reinforcement was fixed above the structural reinforcement of the top slab to provide a continuous electrical path for stray current. Similar mesh in the crossheads was connected to the beams by insulated straps with electrodes embedded in the concrete for periodic checks on passivation.

109. The second stage concreting was usually cast by day 8 and the transfer strength of 36 N/mm^2 was normally achieved three days later. The internal side forms were stripped 12 h after concreting and the internal deck forms were stripped after 24 h. The external side forms were then relased, anchor heads placed, wedges inserted and stressing carried out on day 11. The yard lifting gantry would then transfer the beam to the storage area where grouting of the tendons and finishing work were carried out.

110. A typical 22·3 m long beam would have four tendons of 19/15·7 mm or 27/12·9 mm strand tensioned to jack forces of the order of 3800 kN. Initially 50% of the stressing force was applied using alternate end stressing and average extensions of the order of 150 mm were obtained at 100% load. Multi-strand stressing jacks of capacity 460 t and activated by hydraulic pumps were used for tensioning.

111. After stressing, grouting tubes were fixed at the anchorages and the block-outs cast with grade 45 concrete. The ducts were water flushed and blown through with compressed air before grouting. Ordinary Portland cement grout having a water/cement ratio of 0·45 plus admixture equivalent to 0·2% of the weight of cement was pumped into the ducts with pressure taken to 0·7 N/mm^2. Care was

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

taken to ensure that the grouted ducts were not subjected to shock or vibrations within 24 h of grouting.

Concrete quality control

112. At the outset it was recognized that for the manufacture of such substantial quantities of very large precast beams, close control of concrete production was essential if specification standards were to be met. To achieve this, modern computerized concrete batching plants were erected either in, or close to, all the casting yards and quality control was therefore easily monitored by site staff. Wet-mix batching plants with outputs usually between 50 and 75 m^3/h and equipped with chilling units were generally used. The specified temperature of fresh concrete could not exceed 32°C at time of placing.

113. The benefits of having short travel times between batching and placing of the concrete in the beams showed in good workability and easy vibration, particularly in the highly congested end blocks and webs, resulting in high quality finish after stripping. The methods of placement usually required a pumped mix and most contractors used a super plasticiser having water reducing properties in their concrete mix designs. The pump mixes had target slumps of the order of 120–150 mm and water/cement ratios between 0·38 and 0·4 for the 45 N/mm^2 concrete.

114. An important consideration in the production of precast prestressed concrete beams is to provide a concrete mix that will give high early strengths to allow de-moulding and post-tensioning as soon as possible after concreting. Mix designs aimed to produce strengths of 10 N/mm^2 within 12 h so as to allow side forms to be removed. Final post-tensioning could begin once a transfer strength of 36 N/mm^2 was reached and this was usually achieved between two and four days after concreting. When the usage of moulds was critical and beam cycle times had to be reduced, partial stressing and removal of beams from the moulds was allowed once a strength of 20 N/mm^2 was achieved.

115. The casting yards' batching plants also had to supply the concrete for the prestressed in situ bridges along the railway route and for the construction of crossheads, piers and substructure. For pile and pile-cap construction external concrete supplies, brought in by ready-mix truck, were allowed to supplement the casting yard production.

116. Materials for the production of concrete were tested both by the contractors and by the MRTC's testing laboratory which had been purpose built to test the materials of all MRT contracts. Each contract required strength, grading and chemical tests to be carried out on cement and aggregates, and trial mixes were conducted for all grades of concrete.

117. Throughout the project good quality control was achieved and typical statistical data for the 45 N/mm^2 grade concrete as produced for the 604 phase 1A viaduct beams are given in Table 12 and design mixes in Table 13.

Beam launching gantries

118. A total of 17 launching gantries were used for placing the precast beams for the viaduct contracts. Some of the gantries were reused on contracts which were awarded at later data. Apart from one gantry used on the phase 1A C202 contract, all the launching gantries were conventional, being twin longitudinal girders seated on cross beams with two top travelling hoisting girders for lifting

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

Table 12. Grade 45/P concrete statistical data, phase 1A yards, 1986

Month	Average strength: N/mm²			Standard deviation: N/mm²			Coefficient of variation: %		
	C202	C203	C204	C202	C203	C204	C202	C203	C204
Jan.	56·1	58·5	60·3	3·8	3·7	3·2	6·7	6·3	5·3
Feb.	57·9	54·8	62·7	2·7	4·2	5·9	4·6	7·6	9·4
Mar.	59·6	52·3	62·5	3·4	2·7	3·6	5·7	5·1	5·8
Apr.	60·0	56·7	58·3	3·1	4·5	3·8	5·1	7·9	6·5
May	59·2	56·4	57·3	3·6	3·9	4·6	6·0	6·9	8·0
June	58·5	53·1	56·9	3·6	3·9	3·6	6·1	7·3	6·2
July	58·0	55·5	56·5	3·0	3·8	3·1	5·1	6·8	5·5
Aug.	58·8	57·1	53·0	3·2	3·0	3·6	5·4	5·2	6·8
Sept.	55·3	55·4	55·3	3·0	3·0	3·4	5·4	5·4	6·1

Table 13. Phase 1A yards design mixes (pumped), OPC cement

Contract	Cement: kg	Water: l	Aggregate: kg	Sand: kg	Additive: l/m³	Target slump: mm	Aggregate/ cement ratio	Water/ cement ratio
C202	500	188	950	652	4·1	125	3·204	0·376
C203	425	160	1030	770	5·1	150	4·235	0·376
C204	425	165	1020	770	5·0	120	4·212	0·390



Fig. 28. Launching gantry at Clementi station

the precast beams and lowering them in their final position (Fig. 28). The gantries for phase 1A and 2B viaduct contracts had to lift beams of maximum weight 170 t whereas beams weighing up to 270 t had to be lifted by the gantries on four phase 2A contracts. The C202 contractor designed a single girder, partial steel box and partial box truss, which was supported by two portal frames resting on cross girders.

119. The major problem for the gantry designers was always the very heavy weights of the beams and how to lift and lower them economically and efficiently. They also had to ensure that all equipment was reliable as any prolonged breakdowns were potentially disastrous to the short construction programmes. Another problem area lay in manoeuvring and operating the tight horizontal curves and changes in gradient.

120. The beams were transported from the casting yards on two railcars which travelled on a temporary track along the completed viaduct sections to the launching gantries. The rear railcar was motorized and the beam was seated on a turntable support to assist in the negotiation of horizontal curves. The railcars were powered by diesel generators and they normally travelled at 1 km/h when transporting the beams.

121. The complex operation of launching was carried out with the minimum of disruption to the motorist and general public (Fig. 29). The gantries had to be capable of negotiating horizontal curves of 400 m minimum radius and gradients up to 2·5%. Gantries moved linearly over the pier cross girders either on steel rollers or sliding pads and moved laterally by sliding on the pier cross girders. Hydraulic lifting systems using jacks and prestressing strand were commonly used for lifting and lowering the beams although some gantries used wire rope or chain

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August



Fig. 29. Placing a viaduct beam at Clementi station, launching gantry adjacent to traffic

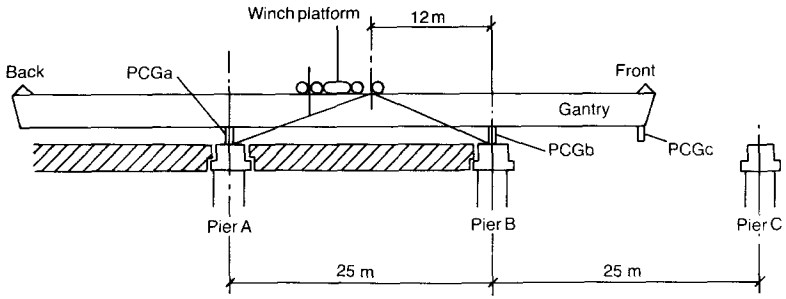
winching systems. On average, it took 4–6 h to launch the gantries onto new spans and 1–2 h to install the beams. This rapid placement of beams made the viaduct sections notable for their speeds of construction as the gantries could comfortably install a beam a day and, when required, two beams a day could be placed. For example, along the phase 1A route 56 beams per month were erected (700 m of twin track viaduct) using the production from four casting yards.

122. The launching gantries either moved forward using winches or, in the case of the Swiss VSL designed gantries, by attachment to the precast beam and then pushing the beam and gantry forward using the motorized railcar. Detailed launching procedures are shown in Figs 30 and 31.

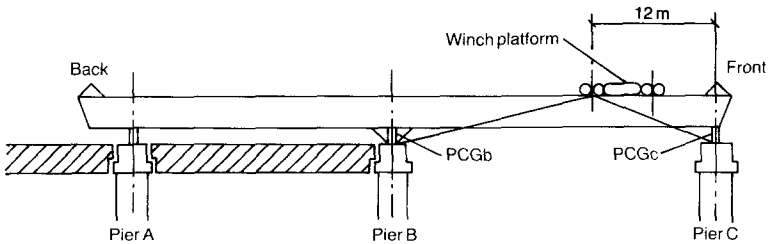
In situ bridges

123. At several locations throughout the MRT system, the standard precast beam viaduct could not be used due to the requirement for long spans over major road, river and railway crossings, or where special structures were necessary at track junctions and cross-overs. A variety of long span structures have been constructed to overcome these problems and Fig. 32 shows a continuous 35 m–55 m–35 m span post-tensioned box girder bridge over a waterway. These bridges were designed to be an extension of the viaduct precast beam form with deeper haunches for additional strength over the longer spans. The 35 m–55 m–35 m spans have six tendons of 37/15·7 mm strand, jack forces of 7550 kN and extensions of about 800 mm. 1000 t capacity multi-strand jacks were used for the stressing and the grouting was usually carried out in the early morning or during the night to avoid the high daytime temperatures.

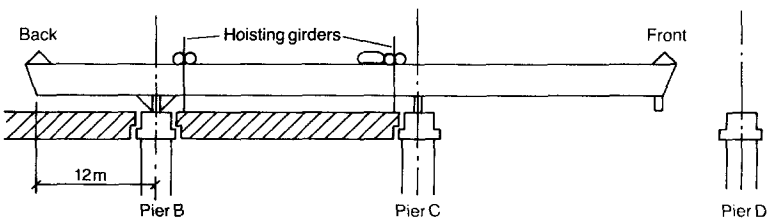
Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August



1. Fix wire ropes from gantry winch platform to pier cross girders (PCG) a and b. Pull in the back rope to winch the launching gantry forward until the front reaches pier C. Install PCGc on crosshead C.



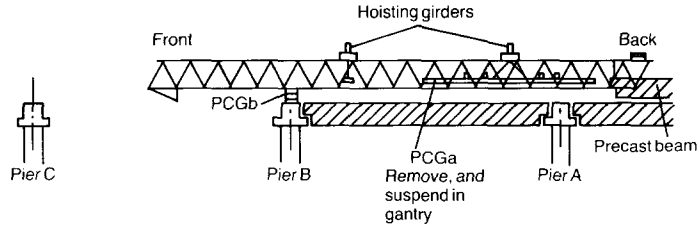
2. Move the launching gantry onto PCGc. Release the wire ropes attached to PCGa and b. Move the winch platform to span piers B and C. Fix the wire ropes to PCGb and c. Pull in the back rope to move gantry forward to its beam launching position.



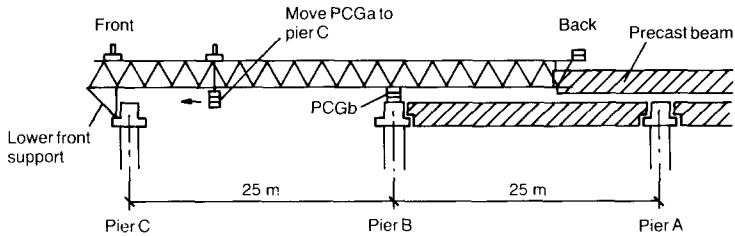
3. Connect precast beam to the front hoisting girder and move the beam into the launching girder. Fix rear hoisting girder to precast beam and move to final position above bearings using the winch. If required, move the launching girder transverlely on PCG b and c using jacks mounted on the cross girder. Lower the precast beam on to the bearings, fix temporary support wedges and grout the bearings.

Fig. 30. Beam launching procedure for winch advanced gantries

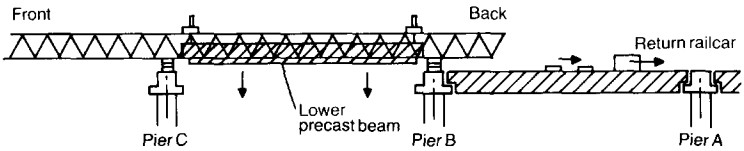
Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August



1. Attach front end of precast beam to rear of launching gantry. Jack up the gantry off the precast beam. Remove pier cross-girder (PCG)a and suspend in the gantry.



2. With the beam acting as a counterweight push the gantry over PCGb, towards pier C using the motorized railcar attached to rear of beam. When the gantry nosing is over pier C crosshead, lower the front supports, jack off crosshead and stabilize gantry. Transport PCGa to a new location on pier C and fix to crosshead with temporary stressing anchors.



3. Advance gantry to final position, release rear of gantry from precast beam. Attach front hoisting girder to front of beam. Move beam into gantry using front hoisting girder together with synchronized pushing by motorized railcar. Release railcar, attach rear hoisting girder to beam, move beam to final position over bearings and lower. Grout bearings with beam remaining suspended in gantry until grout has set.

Fig. 31. Beam launching procedure for a VSL gantry

124. Other span configurations such as 30 m–40 m–30 m, 43 m–43 m, 30 m–40 m–40 m–30 m had similar arrangements of prestressing tendons. Main spans of up to 40 m were also achieved with drop-in precast beams seated on cantilevered side spans. Variations of this form of construction were used on phases 2A and 2B contracts whereas all the phase 1A bridges were continuous structures. Complex structures were also built at the triple track Jurong and Tanah Merah stations where cross-overs and junctions resulted in many tapering bridges, some of multi-cell box girder construction.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

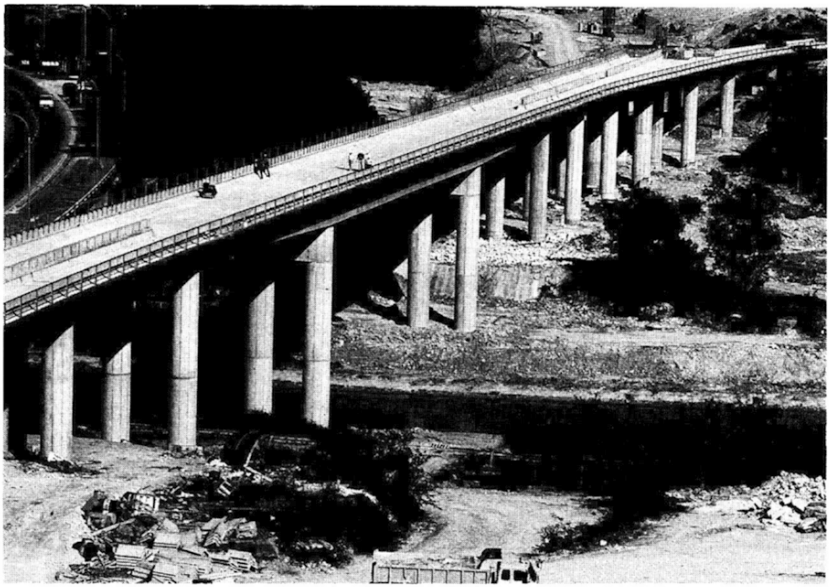


Fig. 32. 35 m–55 m–35 m in situ bridge at Ulu Pandan canal

125. Nearly 3·5 km of viaduct had to be constructed in situ and the temporary works were often complex, especially at the major road junctions where traffic flows had to be maintained and openings had to be left in the falsework for traffic to pass through. This particularly applied to the ten major in situ structures built along the 9 km of phase 1A viaduct which ran for most of its length in the median of a six-lane dual carriageway urban distributor road. Two traffic lanes in each direction had to be maintained throughout the construction period and the in situ bridge sites were very restricted for construction space. At the bridge sites it was also necessary to erect substantial steel and concrete barriers to prevent accidental traffic collision with the temporary works.

126. Also in phase 1A the viaduct passed over the Malayan Railway at two locations and large steel girder falsework trusses were erected over the railway. At both road and rail crossings the falsework was erected at night when traffic was minimal. The longest and highest steel truss erected was at the Ulu Pandan canal crossing, Fig. 32, where the steel truss spanned the full 55 m between piers and was supported off the permanent pile caps by 17 m high falsework towers. After construction of the westbound track prestressed concrete bridge the large steel truss was slid across on top of the falsework towers to construct the eastbound bridge.

127. In situ construction was simplified when the viaduct was off roads so that no traffic openings were required and multiple scaffolding systems could be used to spread the construction loads evenly over the subgrade. Two contracts were notable for the large number of in situ structures to be built. Phase 2A contract 305 and phase 2B contract 403 had a total of 24 in situ bridges to construct

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

between them and substantial plant and labour resources were necessary to meet the tight construction programmes.

128. The timely completion of the in situ bridges often had a critical impact on the precast beams launching programmes. If a launching gantry was unable to travel over the bridges to continue with beam launching, a knock-on effect on beam production could result, as most casting yards could store less than one month's output of beams. All viaduct contracts had key dates for the completion of basic viaduct structure with penalties for delays. Only two of the 13 contractors failed to meet their key dates for viaduct completion and in both cases the track-work contractor was able to assimilate the delays.

Elevated stations

129. Three teams of consultant engineers/architects designed the 26 elevated stations in phases 1 and 2 of the MRT system. Although their designs were evolved from the same set of design criteria and project brief, the consultants produced widely different architectural solutions resulting in interesting contrasts of station styles. All elevated stations have island type platforms with the track separating and passing on either side of the station except for Ang Mo Kio, Jurong and Tanah Merah stations which are triple track stations and have twin island platforms (Fig. 33). The stations are generally 150 m long with platforms between 8·5 and 10·5 m above ground level. The elevated stations are not air-conditioned and the open platforms have large roofs or canopies over the tracks to provide adequate weather protection and ventilation.



Fig. 33. Triple track station at Jurong, phase 2B

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August 2

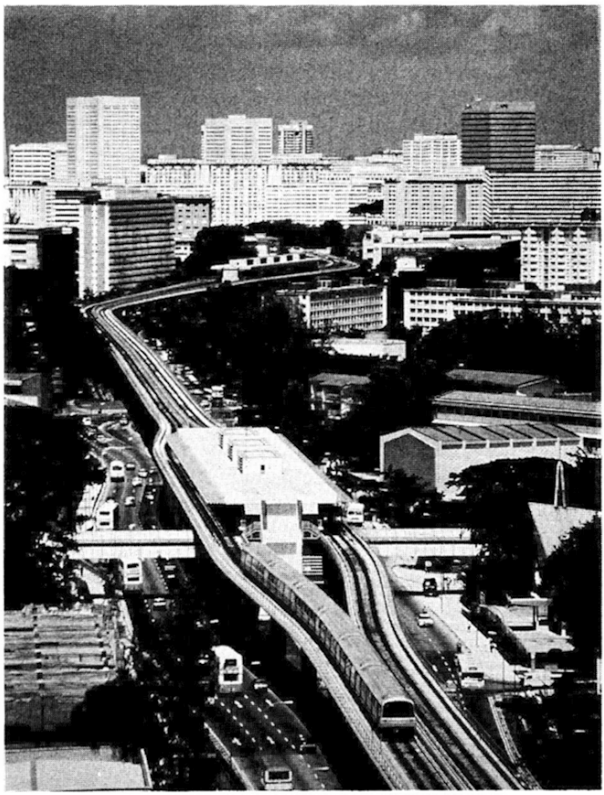


Fig. 34. Commonwealth and Queenstown stations, phase 1A

130. The five phase 1A stations are all similar in design, generally of three levels having plant rooms at ground level, passenger entry/exit gates and the station control room at concourse level, with escalators/stairs to platform level where signalling and telecommunication equipment rooms are located. The station structures are independent of the viaduct, being reinforced concrete framed with reinforced concrete floors and roofs. The stations are noticeable for their glazed pastel coloured tiles and louvre finishes: each station has its own colour for identification. Three of the stations are situated in the median of a six-lane dual carriageway with access to them by footbridges over both carriageways (Fig. 34).

131. The ten phase 2A stations are all two-level buildings consisting of a concourse level usually 1 m above surrounding grade and a platform level 8·5 m above the concourse where the signalling/telecommunications rooms are located. The ticket hall and electrical/mechanical plant rooms are located at concourse level which has been designed as a column-free space with escalators and stairs to the platform above. Eight of the stations are identical and have distinctive 7 m high parabolic arched roof canopies supported by arched ribs, spanning 23 m over the island platform and tracks.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August



Fig. 35. Roof under construction at Aljunied station, phase 2A

132. The station structures are integrated with the viaduct so that portal frames are provided at each viaduct pier onto which span the viaduct beams and also precast prestressed I-shaped platform beams which support the cast in situ platform slab. The 30 t precast platform beams were erected either by the launching gantry or by a purpose designed gantry. The parabolic arched roofs consist of 25 precast concrete ribs at 6·25 m centres, each rib constructed in three segments and jointed on site with bolted steel connections clad with in situ concrete. The ribs were cast in steel moulds either off site or in the viaduct casting yards and erected by crane. They are seated into 120 t precast edge beams erected along each side of the station. Between the ribs a 150 mm thick in situ roof slab was cast using a four-bay, 25 m long travelling form on rails (Fig. 35). The construction cycle time averaged 15 days for a four-bay section. An acrylic polymer gel waterproofing membrane reinforced with glass fibre was applied to the roof concrete after completion.

133. The triple track Tanah Merah station is similar to the other phase 2A stations apart from its arched roof canopy which spans the twin island platforms and has side cantilevers extending over the outside rail tracks. The roof is supported by precast concrete rib frames at 6·25 m centres which are jointed to stub columns located near the centre of each platform. The columns are supported on heavily reinforced longitudinal platform beams. The method of construction was similar to the other stations except that the contractor used conventional scaffolding frames to construct his arched roof slabs instead of a travelling form. The more complicated shape of the roof arch and cantilevers justified this decision and cycle times were comparable to the travelling form construction.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August 2

134. The eight phase 2B stations and the phase 2A Eunos station have structural steel framed roofs which reflect Singapore's cultural heritage in the Chinese, colonial and Malay style roofs that the consultant architects developed. The triple track Jurong station has a space-frame structure which complements a similar space-frame of an adjacent bus transfer centre. Apart from the three-level Jurong station, the phase 2B stations have two levels and most have small ancillary entrance buildings whose roofs match the style of station roofs. Signalling/telecommunications rooms are at platform level and the other electrical/mechanical plant rooms are at ground concourse level.

135. The phase 2B stations, like the phase 2A stations, are also integrated with the viaduct structure with cross-beams seated on corbels cantilevered from the viaduct columns. Two 35 t and two 60 t precast prestressed platform beams span the cross-beams; the heavier platform beams support the roof columns above. Both cross-beams and platform beams rest on rubber bearings and contractors either used the launching gantry or a purpose-made gantry to place them into position. Some contractors preferred to cast the reinforced concrete cross-beams in situ whereas others precast them, together with the platform beams, either in the casting yards or off site. Station architectural finishes included granite tiles for platform and concourse flooring, and polished granite wall finishes and vitreous enamel panelling. The ethnic style roofs were clad in clay tiles apart from the Eunos station Malay roof which has profiled steel roof sheeting.

136. The electrical and mechanical items in the stations were designed and installed by separate system-wide contractors. The civil contractors were responsible for the co-ordination and interaction of these contractors to ensure that their requirements were met. This important co-ordination role required each civil contractor to produce combined services drawings showing the major cable routes and main items of plant together with details of structural openings and embedded items. The co-ordination work often resulted in considerable structural implications due to civil contractors having to accommodate the on-going amendments or additional requirements of the electrical and mechanical contractors which usually took a long time to finalize. The complexity of the project and the importance of ensuring that both civil and system-wide contractors' work was executed within strict time limitations required that the civil contractors produced detailed combined installation programmes for electrical and mechanical construction installation and architectural finishes. In particular, detailed degrees of completion for specific plant rooms had to be met by the civil contractors before the system-wide contractors could enter to carry out their works.

Track laying

Phase 1

137. To meet the very tight equipping schedule provision was made to lay track in Bishan depot starting in September 1985, although much of the depot construction was incomplete at that time. The depot then served as the supply site, with work proceeding southward into the tunnels and northward onto the viaduct to the phase 1 terminus at Yio Chu Kang. Provision was also made at the cross-over west of Outram Park to keep the cut-and-cover box roof open as access for track laying eastwards and northwards; all materials including works trains were lowered into the tunnel. Additional access was made available at intermediate points for the introduction of materials, particularly those for the floating slab

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

track. Delivery pipes from ground level to the running tunnels were provided in each section between stations to enable concrete to be brought directly from the surface near placing points rather than having to bring it in by train from the main depot area. As work progressed and electrical and mechanical equipping began, it became necessary to integrate the works train schedule for both system-wide contractors and the track laying. Eventually this meant that the track-work contractor was limited to one train movement per track per 12 h shift. To mitigate this restriction it was necessary for the contractor to adopt 24 h working. Careful marshalling of works trains was obviously essential to avoid delays. The contractor's principal items of plant were: seven locomotives (five diesel and two battery powered); four track laying gantries; 28 flat wagons; 12 ballast wagons; three rail mounted concrete agitators; one ballast tamping machine; one ballast regulator; and one fixed monorail gantry.

138. Before track laying could start a detailed wriggle survey of the tunnels and viaducts was carried out by the main civil contractors and agreed by the MRTC. In the tunnels seven points around the circumference were surveyed to establish the clearance from the structure gauge. This was typically carried out every 2·5 m in cut-and-cover tunnels and every 3 m (i.e. third ring) in bored tunnels, except at tight spots where measurements were taken every 1 m. Viaduct clearances were similarly established.

139. To avoid rebuilding, modifications to the theoretical alignment were made, provided that there was no reduction in clearances. No rebuilding has been necessary to date.

Track laying in tunnels

140. *Slab track.* The majority of the track in the tunnels is formed by concrete sleepers set in concrete. The fixed monorail gantry at the yard was used to load eight prefabricated 18 m long track panels onto a specially equipped monorail gantry train. In the tunnel the panels were off-loaded from the train by the track-laying gantry. The gantry was supported at the rear on the previously laid track and at the front on the concrete tunnel invert placed by the main civil contractors, Fig. 36. Panels were temporarily fish-plated together and then the works train pushed the gantry forward to repeat the cycle.

141. The track panels were supported off the tunnel base concrete by sacrificial concrete blocks located below the rails between sleepers. The final rail level was obtained by means of small metal spindles with a screw adjustment set on top of the blocks; these blocks were removed after the in situ concrete around the sleepers had set. Horizontal alignment was obtained by the use of adjustable steel props. Thermit welding was used to provide continuously welded track throughout. Slab reinforcement was placed and concreting carried out in successive sleeper bays either by concrete pump or rail mounted agitator. Traffic was not permitted over freshly placed concrete for seven days, so work had to alternate between tunnels to avoid delays. The average rate of advance achieved so far is 70 m/day.

142. *Floating slab track.* Following the completion of the wriggle survey, compensating concrete edges were laid against the varying tunnel walls to allow a standard length concrete unit to be employed. A purpose built frame was used to set the bearings into position, Fig. 37. Both line and level tolerances were set at ± 2 mm. The bearings were grouted into position with a non-shrink grout. The maximum number of bearings placed in a 24 h day was 200.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August

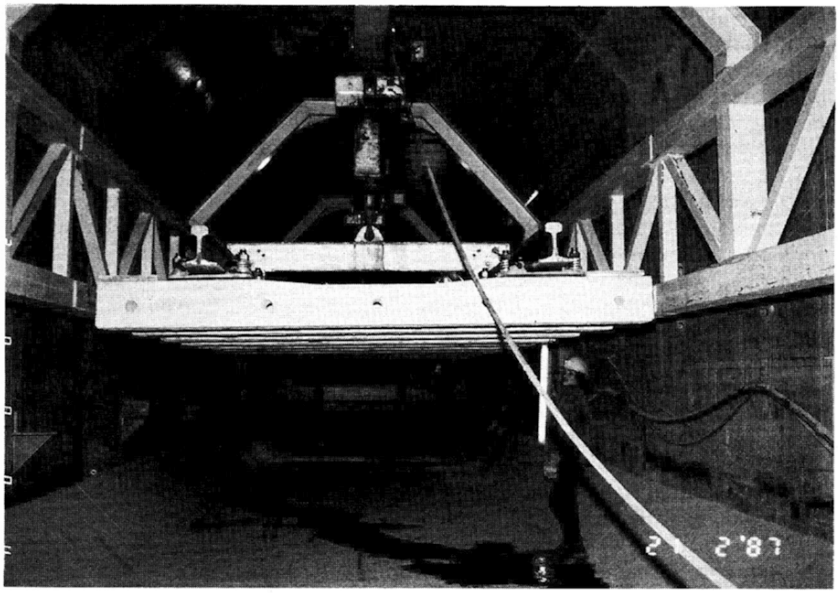


Fig. 36. Laying slab track panels by gantry crane in a cut-and-cover tunnel



Fig. 37. Positioning floating slab track units on elastomeric bearings

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August



Fig. 38. Viaduct track laying by gantry crane

143. The works train would carry in 24 concrete units, and a track laying gantry unloaded the concrete units and placed them on the rubber bearings. Concrete sleepers were placed in recesses in the tops of the units and rails laid and fastened to the sleepers. The track was finally aligned with specially designed screw plates and adjustable props. Sleepers were concreted into the units using a non-shrink concrete. A maximum of 78 units have been placed in a 24 hour day. Overall the floating track slab has been laid at an average of 20 m/day.

144. *Points and crossings.* For points and crossings the work proceeded conventionally in fixed slab track lengths with the base plates used as templates, and all bolt holes drilled into the concrete slab. The pads were secured by glue to the baseplates during grouting. Where points and crossings were required within floating slab track areas, they are located on a floating track bed consisting of an 80 mm thick precast filigree slab supported off the bearings and on which a heavily reinforced concrete slab 450 mm thick is cast.

Track laying on viaducts

145. End tipping lorries ran on the protective screed to the waterproofing to lay a 200 mm thick bottom ballast layer. Track was laid either by track laying gantry (Fig. 38) or a Hiab crane arm mounted on the works train. Ballast laying to the full depth of 430 mm was carried out by ballast wagons running over the track, and then a Plasser and Theurer tamping machine (type 07-275) was used to bring the track to its final line and level. Output has averaged 500 m/week (80 m/day).

146. Points and crossings on the viaducts were pre-assembled, transported to site by works train and then placed by the Hiab crane.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August 2

Rail welds

147. All welding was carried out in situ using the Thermit SKV Alumino Thermit process. Initially local welders were trained by specialist personnel provided by Thermit Inc, UK. Continuous welded rail was formed by releasing sections of rail (up to 1000 m long) from their fastenings and then welding them together at the neutral temperature 40·5°C in tunnels and 43°C in the open with a tolerance of ± 3°C. Rail tensors were used to draw rail sections together to simulate the stress condition at the neutral temperature if it became necessary to carry out work at other temperatures. Six welds per team have been achieved per shift. All welds were checked ultrasonically before acceptance.

Installation of third rail

148. After track laying was complete third rail support brackets were screwed to the sleepers at intervals not exceeding 6·3 m. Wooden sleepers were drilled in situ but a cast-in plastic insert was routinely provided in the concrete sleepers. A purpose made glass fibre polyester insulator was then fixed.

149. The composite aluminium steel conductor rail was clipped by a simple claw mechanism and the 15 m lengths fish-plated together. Cable fixing to the conductor rail and cross bonding of the running rails followed. A protective cover to the top and side of third rail was finally installed.

Conclusions

150. Of the ten stages for public opening of the Singapore MRT system the five which have already opened have done so ahead of programme. It is expected that the remaining stages will open ahead of their already accelerated completion dates. The overall cost of the system is expected to be almost exactly that forecast five years ago. These facts would represent a significant achievement in any country, let alone one with little previous experience of building railways or of tunnelling.

151. Much of the credit of the achievement must go to those most directly involved in the construction, the contractors and consultants, and the MRT staff who co-ordinated and inspected the work. However, it would have been impossible without the clear vision of the Singapore Government which resulted in meticulous planning long before construction started. The whole-hearted co-operation of many government departments was also absolutely vital to the success of the project.

References

1. POH K. B. *et al.* Some MRT experiences of the soils and geology of Singapore. *Proc. Singapore Mass Rapid Transit Conf., Singapore, 6–9 April 1987.*
2. PECK B. *et al. Foundation engineering.* Wiley, New York, 1974, 2nd edn.

Downloaded from www.emerald.com/jpric/article-pdf/86/4/709/2684428/iicep_1989_2628.pdf by National University of Singapore user on 18 August